LEBANESE UNIVERSITY
FACULTY OF ENGINEERING III

Mechanical Engineering

SENSORS

GREENHOUSE MONITORING AND AUTOMATIC CONTROL SYSTEM

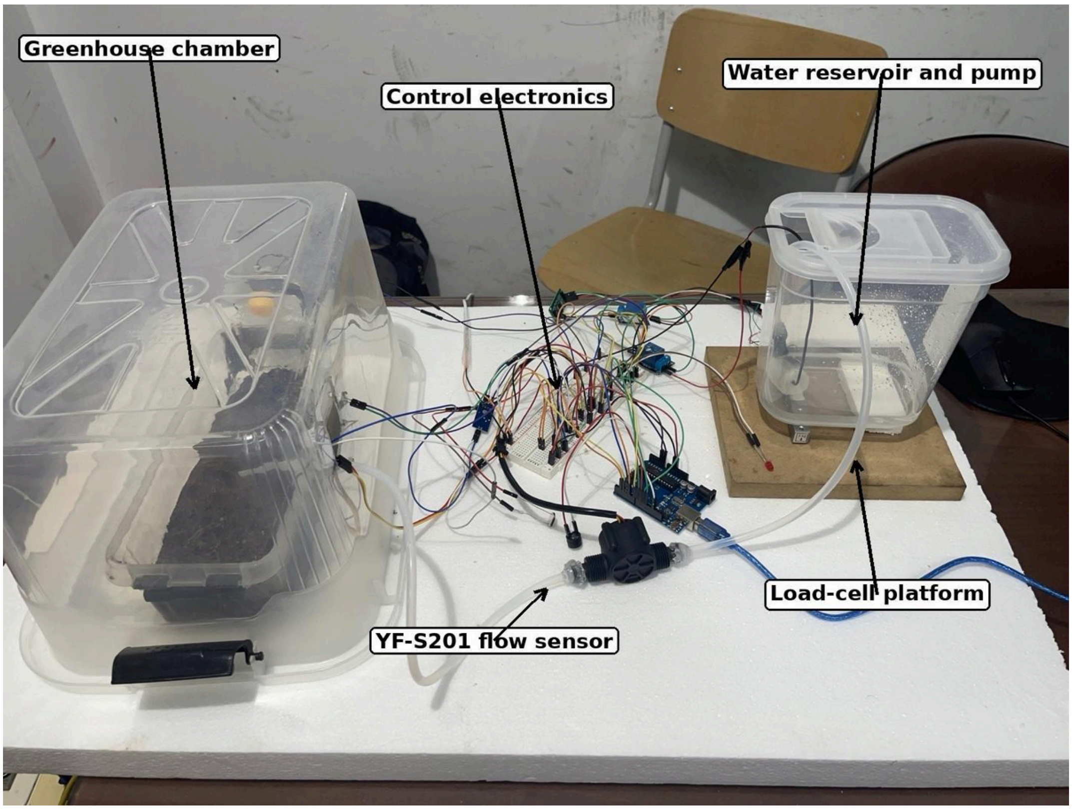


Figure 1. Final integrated smart greenhouse prototype.

Document	Final Technical Report
Prepared by	Walid Al Basha - 6726 Mohammad Fakih - 6828 Nancy Mezher - 6814 Ahmad Ghabris - 6866
Course	Sensors
Instructor	Dr. Mohamad Aoude
Submission date	July 2026

Abstract

This project presents the design and implementation of an Arduino-based smart greenhouse that monitors the main environmental and water-related variables affecting a small growing enclosure. The prototype combines five required sensing modalities: optical measurement using a light-dependent resistor, force measurement using a load cell and HX711 interface, liquid flow measurement using a YF-S201 Hall-effect flow sensor, and temperature and humidity measurement using a DHT11 sensor. A resistive soil-moisture probe was added because soil condition is the direct variable used to decide when irrigation is required. Two automatic responses were implemented. A water pump is switched through a relay when the soil becomes dry, while a small ventilation fan is activated when the air temperature rises above the selected limit. The reservoir is weighed continuously so that irrigation can be inhibited when the available water becomes too low, and the flow sensor confirms that water is actually moving when the pump is energized.

The work included component selection, electrical integration, firmware development, calibration, functional testing and analysis of measurement limitations. The final assembly uses an Arduino Uno, two relay modules, a submersible DC pump, tubing, a transparent greenhouse chamber, a separate water reservoir and breadboard-based signal wiring. The firmware reads the sensors at suitable intervals, averages the load-cell signal, counts flow pulses using an interrupt, applies calibration equations, controls the actuators with threshold hysteresis, and sends formatted measurements to the serial monitor. A representative sixty-second dataset is used to demonstrate how the pump, fan, soil moisture, water flow and reservoir mass interact during operation. The results show that the architecture can perform continuous monitoring and basic closed-loop control using low-cost hardware. The project also demonstrates the practical importance of calibration, sensor placement, electrical isolation, common grounding, noise reduction and safe separation between water and electronics.

Table of Contents

List of Figures

List of Tables

Abbreviations and Symbols

Symbol	Meaning
ADC	Analog-to-digital converter
CSV	Comma-separated values
DHT	Digital humidity and temperature sensor family
GPIO	General-purpose input/output
HX711	24-bit load-cell analog front end and ADC
LDR	Light-dependent resistor
NO / NC	Normally open / normally closed relay contact
RH	Relative humidity
Q	Volumetric flow rate, L/min
f	Pulse frequency, Hz
m	Mass, kg
T	Temperature, °C

1. Introduction

1.1 Background

A greenhouse creates a controlled space around plants, but the enclosure does not control itself. Temperature can rise because solar radiation and electrical equipment add heat. Relative humidity can change as water evaporates from the soil. The soil may dry even while the air still feels humid, and a small reservoir may empty without being noticed. In a manually operated setup, a person must repeatedly inspect several conditions and decide whether to ventilate or irrigate. This approach is acceptable for a few plants, but it is inconsistent and wastes time and water.

A smart greenhouse replaces repeated manual checking with sensors, a controller and actuators. The controller does not simply collect numbers. It interprets the measurements and performs useful actions. In this prototype, soil moisture determines the need for irrigation, temperature determines the need for ventilation, the reservoir mass indicates whether enough water remains, and the flow sensor confirms that the irrigation path is functioning. Light and humidity provide additional environmental information for the operator.

The course project required the integration of optical, force, flow, temperature and humidity sensing into one Arduino-based system. It also required calibration, continuous acquisition, threshold detection, an automatic response, real-time visualization and a minimum sixty-second demonstration [1]. The smart greenhouse scenario was selected because each required sensor has a clear engineering purpose rather than being added only to satisfy the component count.

1.2 Project Aim

The aim was to build a low-cost prototype that can measure the condition of a small greenhouse and react automatically to dry soil and high temperature. The system was designed as an educational engineering prototype rather than a commercial agricultural product. The priorities were understandable wiring, visible operation, simple calibration, safe low-voltage switching, and enough modularity to allow each sensor to be tested independently.

1.3 Objectives

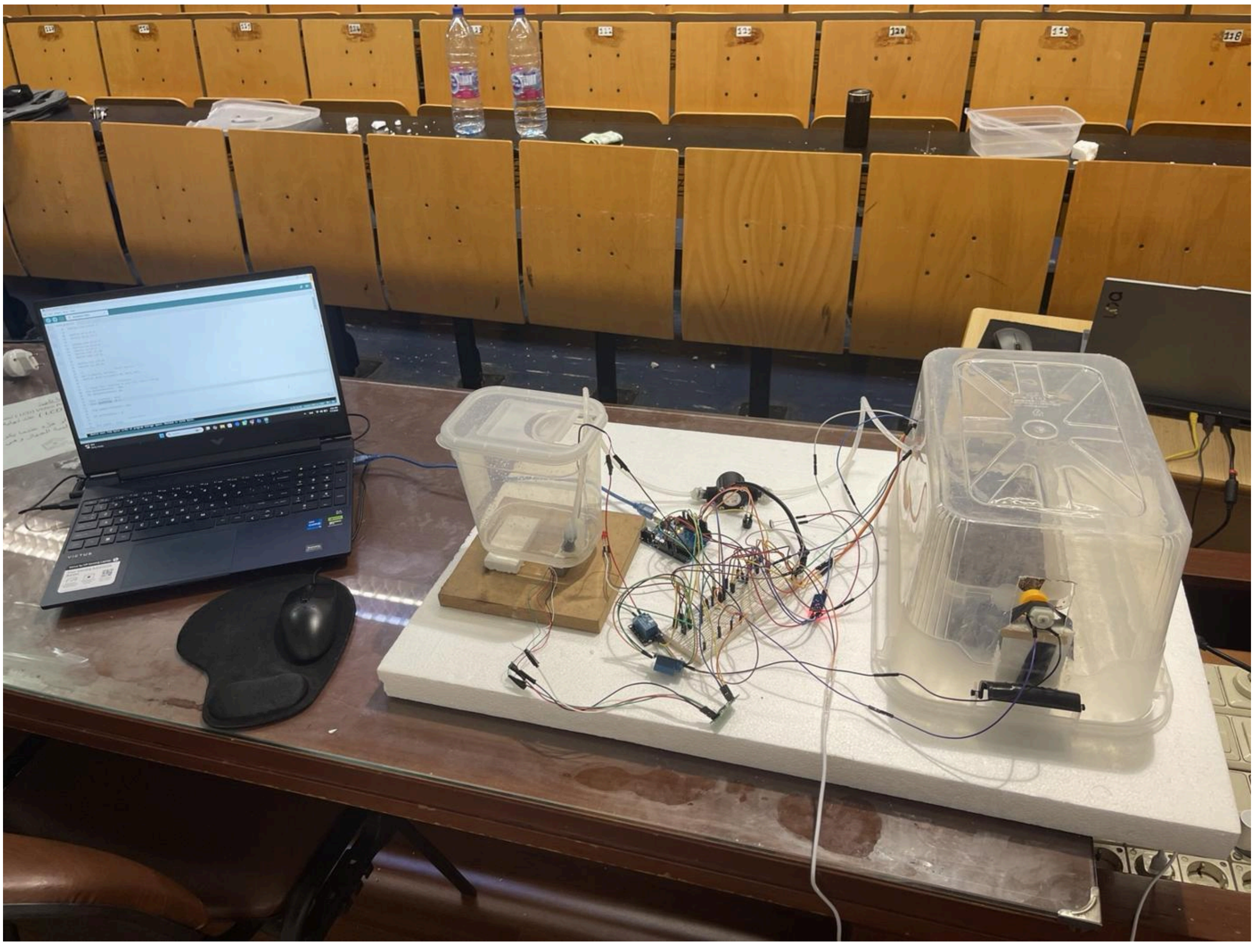
Table 1. Main project objectives and verification method.

Objective	Verification method
Measure greenhouse air temperature and relative humidity	DHT11 readings displayed through the serial interface
Estimate soil wetness	Analog soil-moisture signal converted to a calibrated moisture index
Monitor the available water	Reservoir mounted on a load cell and read through HX711
Measure delivered water flow	YF-S201 pulse frequency converted to L/min
Monitor light level	LDR voltage-divider output converted to an illuminance estimate
Irrigate automatically	Relay-driven pump turns on below the dry-soil threshold
Ventilate automatically	Relay-driven fan turns on above the high-temperature threshold
Detect abnormal operation	Low tank mass and pump-without-flow conditions are identifiable
Record and visualize data	Timestamped serial output suitable for CSV logging and plotting

1.4 Scope and Limitations

The prototype monitors one soil container inside one plastic enclosure. It does not model plant species, root depth, solar heat gain or nutrient concentration. Irrigation is controlled by a single soil-moisture measurement, so the reading represents the region around the probe rather than the entire container. The fan is operated by on/off control rather than variable-speed control. The reservoir measurement assumes that one kilogram of mass loss is approximately one litre of water delivered. These simplifications are appropriate for a three-week course project, but they are discussed later as sources of uncertainty and opportunities for future work.

The photographs show a breadboard implementation with jumper wires and open relay modules. This was suitable for a laboratory demonstration, but a permanent greenhouse installation would require an enclosure, strain relief, sealed connectors and a printed circuit board or terminal-based wiring.



2. Problem Statement

2.1 Engineering Problem

The problem is to maintain acceptable conditions in a small greenhouse without continuous human supervision. Three practical failures are especially important. First, dry soil can stress the plant if irrigation is delayed. Second, temperature can increase inside the enclosure because the plastic walls restrict natural air exchange. Third, a pump command does not guarantee that water reaches the soil: the reservoir may be empty, the tube may be blocked, or the pump may be disconnected. A useful system therefore needs both environmental sensing and water-path verification.

The controller must make decisions from imperfect measurements. Soil probes drift with corrosion and soil composition. DHT11 measurements are quantized and relatively slow. LDR response is nonlinear. Flow pulses fluctuate at low flow, and a load cell is sensitive to mechanical contact and cable forces. The design problem is therefore not only to connect sensors, but to select sampling intervals, establish calibration equations, filter noise, define thresholds and keep the actuators in a safe state when a reading is invalid.

2.2 Functional Requirements

Table 2. Functional requirements derived from the course proposal.

ID	Requirement	Implementation in this project
FR-1	Acquire all required sensor modalities	Temperature, humidity, light, force/weight and flow are read during one program run.
FR-2	Measure soil condition	A separate analog soil probe supplies the irrigation decision variable.
FR-3	Automatic irrigation	Pump is energized when soil is dry and water is available.
FR-4	Automatic ventilation	Fan is energized when temperature exceeds the high limit.
FR-5	Continuous update	The display must continue updating rather than stopping inside a blocking routine.
FR-6	Flow verification	Flow pulses are measured while the pump operates.
FR-7	Reservoir supervision	Tank mass is measured and negative or unstable values are corrected by tare and calibration.
FR-8	Real-time display	Serial output provides timestamped system status.
FR-9	Demonstration duration	System can operate continuously for at least 60 seconds.
FR-10	Low-voltage safety	No direct mains connection; maximum actuator supply does not exceed 12 V DC.

2.3 Design Specifications

- All sensor and controller electronics operate from 5 V logic.
- Pump and fan power are switched through relay contacts rather than directly from Arduino pins.
- The control loop remains responsive while the HX711 and DHT11 are read.
- The flow sensor is connected to an interrupt-capable pin so that pulses are not missed.
- The pump is prevented from operating below a minimum reservoir mass.
- Hysteresis is used for pump and fan control to avoid rapid relay switching near a threshold.
- Measurements are printed in a consistent order that can be copied into a CSV file.
- Water containers and tubing are physically separated from the breadboards and USB connection as far as practical.

2.4 Success Criteria

Success is demonstrated when dry soil starts irrigation, the flow sensor detects water, soil moisture rises, and the pump stops at the wet threshold. High temperature must start the fan, while cooling below the reset threshold must stop it. The tared load-cell output must remain non-negative, and the serial data must refresh throughout the demonstration.

3. System Architecture

3.1 Overall Architecture

The system has four layers: sensing, processing, actuation and visualization. The sensing layer contains the DHT11, LDR, soil-moisture probe, load cell with HX711, and YF-S201 flow sensor. The processing layer is the Arduino Uno. The actuation layer contains two single-channel relay modules, a submersible pump and a small DC fan. The visualization layer is the USB serial connection to the Arduino Serial Monitor or Serial Plotter.

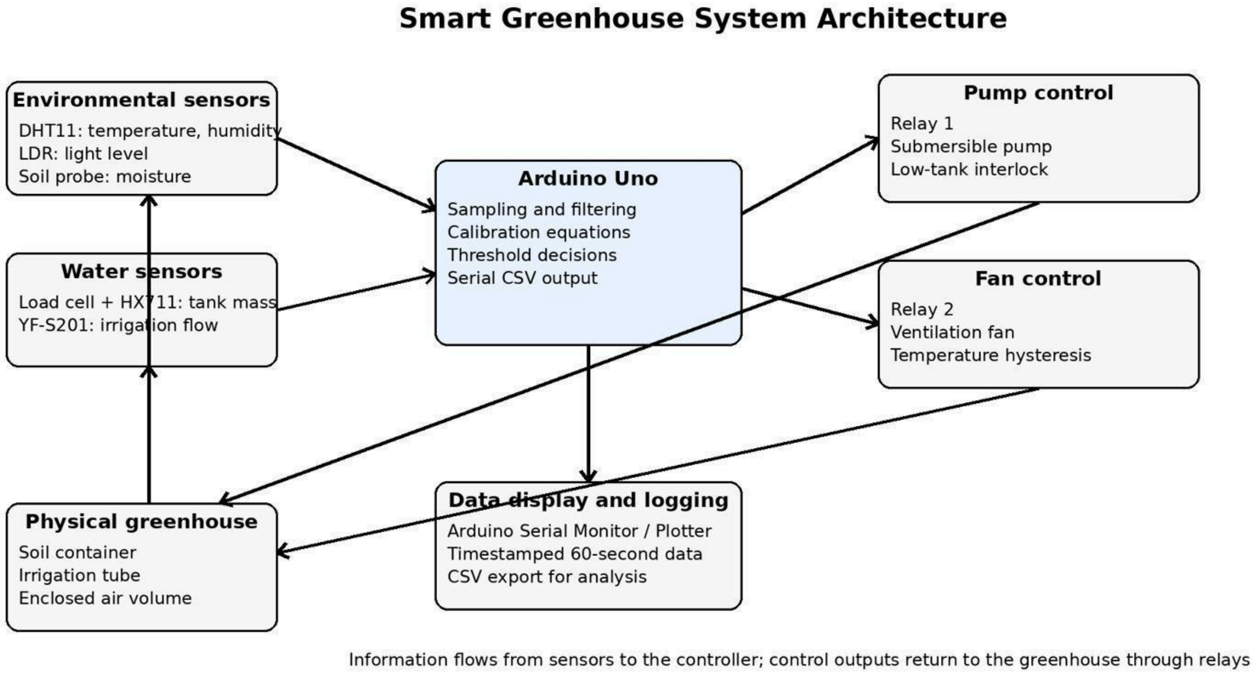


Figure 2. System-level architecture and information flow.

The Arduino Uno was suitable because it provides six analog inputs, fourteen digital I/O pins and a 16 MHz ATmega328P processor, which is sufficient for the selected sensors and two relay outputs [2]. The design uses analog inputs for the soil probe and LDR, digital timing for the DHT11 and HX711, an interrupt input for the flow pulses, and digital outputs for the relays.

3.2 Physical Arrangement

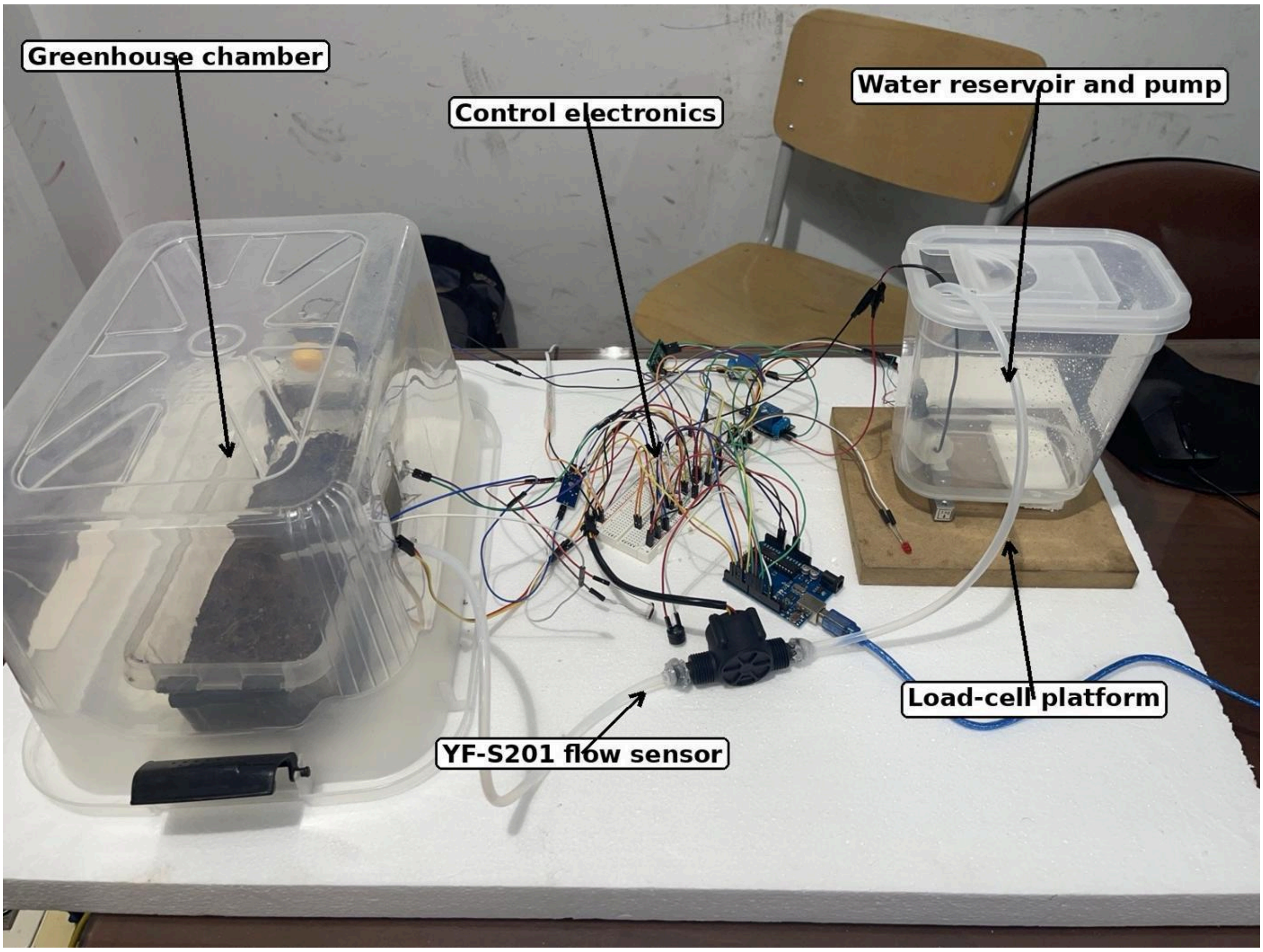


Figure 3. Final prototype with the main subsystems identified.

The greenhouse chamber is located on one side of the base and the water reservoir is located on the other. The electronics are kept between them so that the signal wires remain short, while the tubing forms a separate hydraulic path. The transparent chamber contains the soil container, DHT11, LDR, soil probe, irrigation outlet and fan. The reservoir contains the submersible pump and rests on the load-cell platform.

3.3 Closed-Loop Irrigation Sequence

1. The Arduino reads the soil-moisture index and reservoir mass.
2. If the soil value is below the dry threshold and the tank mass is above the minimum safe value, the pump relay is energized.
3. Water leaves the reservoir, passes through the YF-S201 flow sensor and enters the greenhouse irrigation tube.
4. The flow sensor produces pulses that confirm water movement and provide a flow-rate estimate.
5. The soil reading increases as water reaches the region around the probe.
6. When the wet threshold is reached, the pump relay is released.
7. The controller continues measuring the system, so another irrigation cycle can begin later if the soil dries.

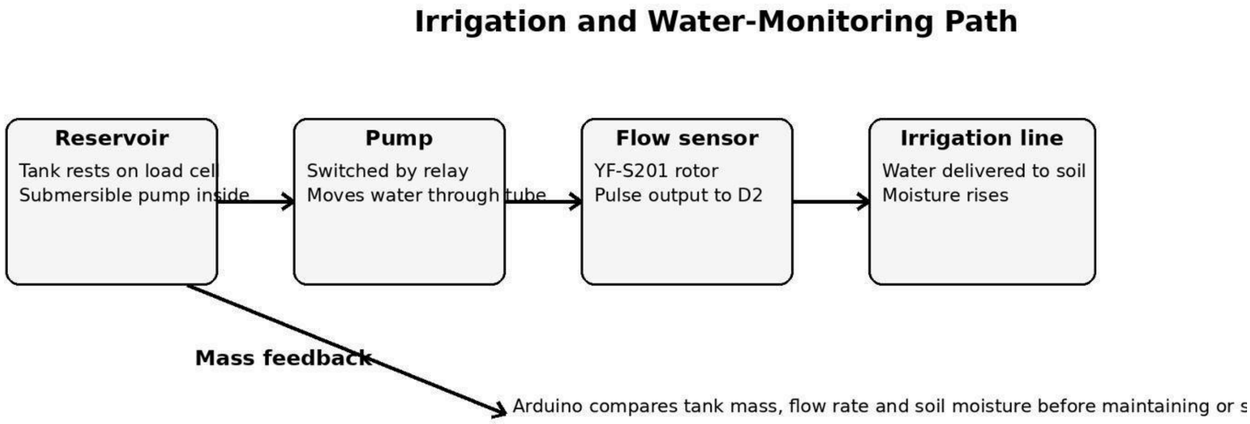


Figure 4. Irrigation and water-monitoring path.

3.4 Closed-Loop Ventilation Sequence

The temperature loop is simpler. The DHT11 provides the greenhouse air temperature. When the temperature exceeds the upper threshold, the fan relay is energized. The fan exchanges air through the openings in the chamber. When the temperature falls below the lower reset threshold, the fan is stopped. The difference between the ON and OFF thresholds is hysteresis; it prevents the relay from repeatedly clicking when the measured temperature fluctuates around one value.

3.5 Data Flow and Visualization

Each output line contains elapsed time, calibrated sensor values and actuator states. This format supports three uses at once: the operator can read the current state, the Arduino Serial Plotter can display trends, and the text can be saved as a CSV file for Excel, MATLAB or Python analysis. The proposal required timestamped logging and real-time visualization [1]. The prototype uses elapsed milliseconds as a relative timestamp because no real-time clock module was included.

4. Sensor Selection and Justification

Table 3. Sensor and actuator selection summary.

Function	Selected device	Measured quantity	Justification
Temperature and humidity	DHT11	0-50 °C and typical greenhouse RH range; digital output	One device supplies two required modalities; low cost and simple wiring.
Optical	LDR in voltage divider	Relative light level / estimated lux	Simple analog sensor with strong response to changes in illumination.
Soil condition	Resistive soil-moisture probe	Relative soil wetness	Direct variable for irrigation control and easy demonstration by adding water.
Force / weight	Load cell + HX711	Reservoir mass	Measures water availability without placing electrodes inside the tank.
Flow	YF-S201	Water flow rate in L/min	Pulse output gives independent confirmation that the pump is delivering water.
Actuator switching	5 V relay modules	Pump and fan ON/OFF	Allows Arduino logic to switch separate DC loads.
Controller	Arduino Uno	Acquisition and control	Enough analog/digital pins, USB serial and wide library support.

4.1 DHT11 Temperature and Humidity Sensor

The DHT11 combines a humidity-sensitive element, a temperature sensor and internal signal processing. It provides calibrated digital values through a single data line. The device is inexpensive and adequate for a demonstration in the normal indoor temperature range. Its main limitations are coarse one-degree and one-percent output resolution, relatively slow sampling, and typical accuracy around ±2 °C and ±5 %RH [3]. For this reason, the firmware does not read it as quickly as the analog sensors.

The DHT11 is positioned inside the greenhouse but away from direct water spray and away from the fan outlet. This placement is important. A sensor immediately beside the fan would respond to the incoming room air rather than the average enclosure condition, while a wet sensor body would produce misleading humidity readings.

4.2 Soil-Moisture Sensor

The resistive soil probe consists of two conductive electrodes. The electrical resistance between them decreases as the water content and ionic conductivity of the soil increase. The associated module converts this resistance change into an analog voltage. In the wiring used for this prototype, the calibrated percentage is derived from the ADC count rather than relying on the module digital comparator.

The probe was selected because it directly supports the irrigation decision and visibly responds during the demonstration. The main drawback is electrode corrosion and dependence on soil composition. The probe should therefore be powered only when a reading is needed in a permanent design, or replaced by a capacitive sensor. SparkFun describes the same basic principle: the two probes act as a variable resistor and wetter soil gives better conductivity [4].

4.3 LDR Light Sensor

An LDR changes resistance with incident light. It is connected as part of a voltage divider so that the Arduino reads a light-dependent voltage. The response is nonlinear, so a power-law calibration gives a more useful illuminance estimate than a direct linear conversion. The project does not control an artificial lamp; light is monitored and classified as dark, moderate or bright. Arduino documentation uses the same voltage-divider method for connecting an LDR to an analog input [5].

4.4 Load Cell and HX711 Interface

The load cell is a strain-gauge beam that produces a very small differential voltage when bent by the reservoir weight. This signal cannot be read directly with good resolution by the Arduino ADC. The HX711 provides a high-gain differential front end and a 24-bit converter intended for bridge sensors.

The Adafruit implementation used in the updated project code communicates through a data line and clock line; the preserved final assignments are HX711_DT = 4 and HX711_SCK = 5 [6].

The reservoir is weighed rather than using a float switch because mass provides a continuous quantity. With water density close to 1 kg/L at room temperature, a change in tank mass also provides an independent estimate of delivered volume. We had an early problem with negative values. This was addressed by taring the empty platform, maintaining a consistent load direction and applying a calibration factor with the correct sign.

4.5 YF-S201 Water-Flow Sensor

The installed black YF-S201 contains a rotor and Hall-effect sensor. Water rotates the rotor and the sensor produces digital pulses. The label on the installed device identifies a nominal 1-30 L/min range and 5-24 V DC operating supply. The standard relation is approximately $f = 7.5Q$, where f is pulse frequency in hertz and Q is flow rate in litres per minute, although individual calibration improves accuracy [7].

The flow sensor is placed after the pump and before the irrigation outlet. This makes it a functional check. If the pump relay is ON but the measured flow remains near zero, the system can flag an empty reservoir, blocked tube, disconnected pump or failed sensor.

4.6 Relays, Pump and Fan

The Arduino cannot directly supply the current required by the pump or fan. Each actuator is therefore switched by a 5 V relay module. The control pins marked S, + and - connect to the Arduino signal, 5 V and ground. The screw terminals are COM, NO and NC. For a normally-off actuator, the external supply path uses COM and NO, so the pump or fan receives power only when the relay is energized.

Two separate relays were used so irrigation and ventilation can operate independently. The actuator supply is kept below the 12 V limit stated in the proposal.

5. Hardware Design and Schematics

5.1 Greenhouse Chamber

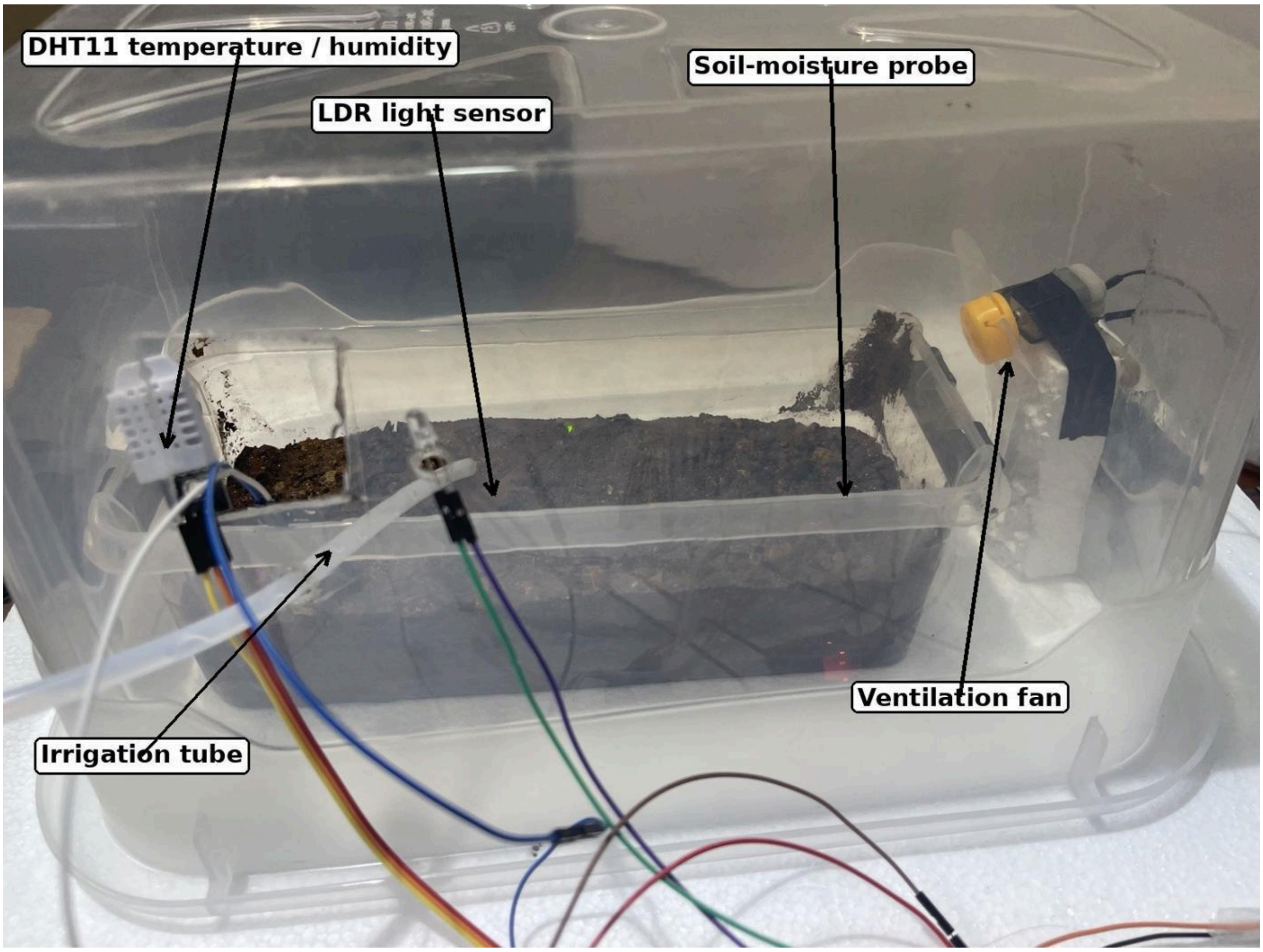


Figure 5. Greenhouse chamber and sensor placement.

The greenhouse chamber is a transparent plastic box containing a smaller soil tray. Openings were cut for the fan, irrigation tube and wires. The DHT11 is suspended above the soil, the LDR faces the enclosure interior, and the soil probe is inserted into the root-zone region. The small fan is fixed to a side opening so that it exchanges air rather than only circulating air inside a sealed box.

The irrigation tube is supported so that it delivers water near the soil surface without contacting the sensor wiring. This arrangement also allows the operator to see water movement during the live demonstration.

5.2 Reservoir, Pump and Load Cell

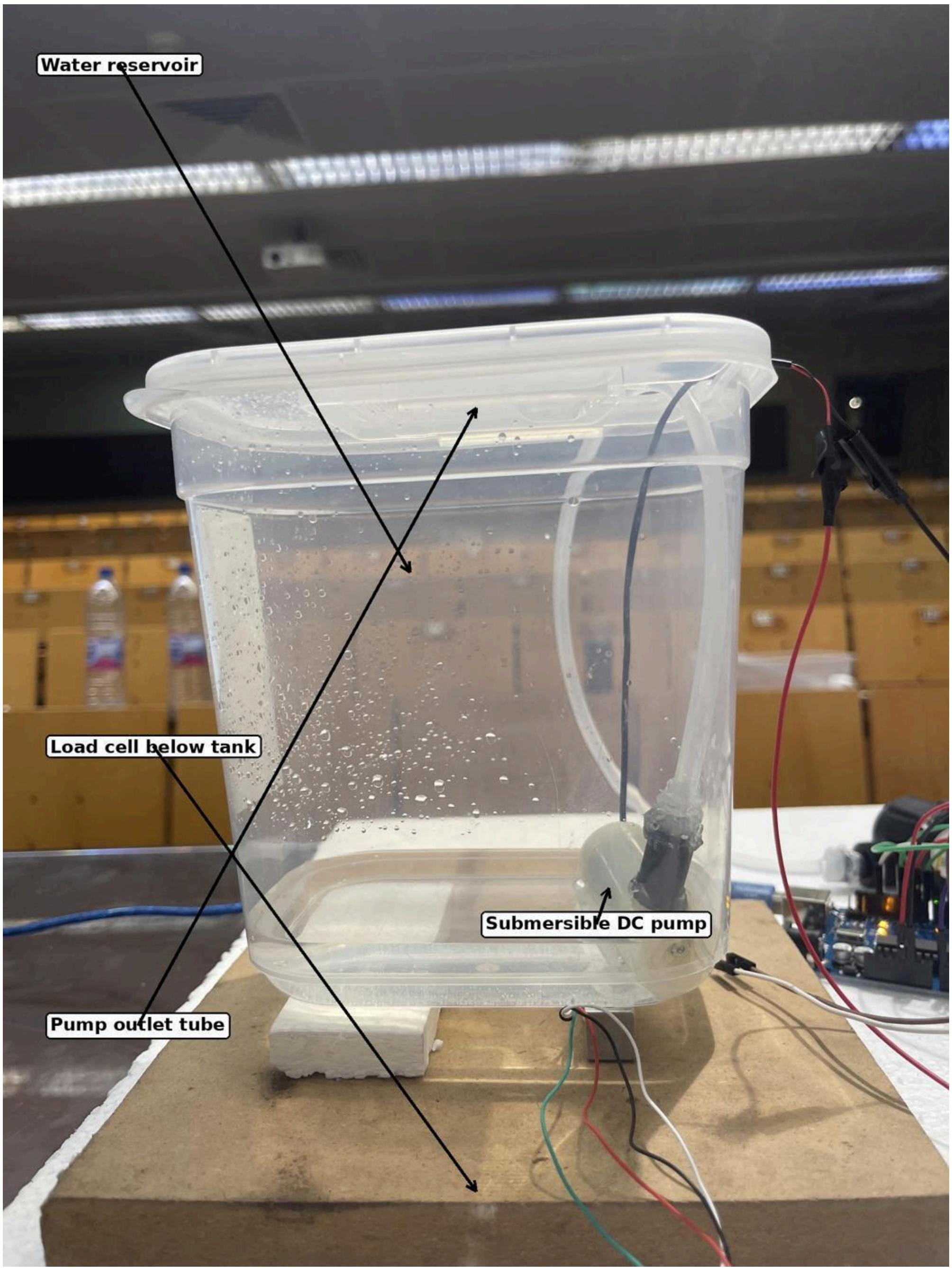


Figure 6. Reservoir, pump and load-cell arrangement.

The submersible pump is located at the bottom of a separate transparent reservoir. The outlet tube rises through the lid and connects to the flow sensor. The reservoir rests on a platform supported by the load cell. Only the tank and its contents should load the sensing platform. A tube pulling sideways or a cable touching the base can change the reading, so the tubing is left with slack.

The pump power wires leave the reservoir above the water line and are joined to the relay-switched DC supply. The low-voltage supply and relay contact wiring are kept outside the wet area. The load-cell wires connect to the HX711 excitation and differential input terminals.

5.3 Electronics Assembly

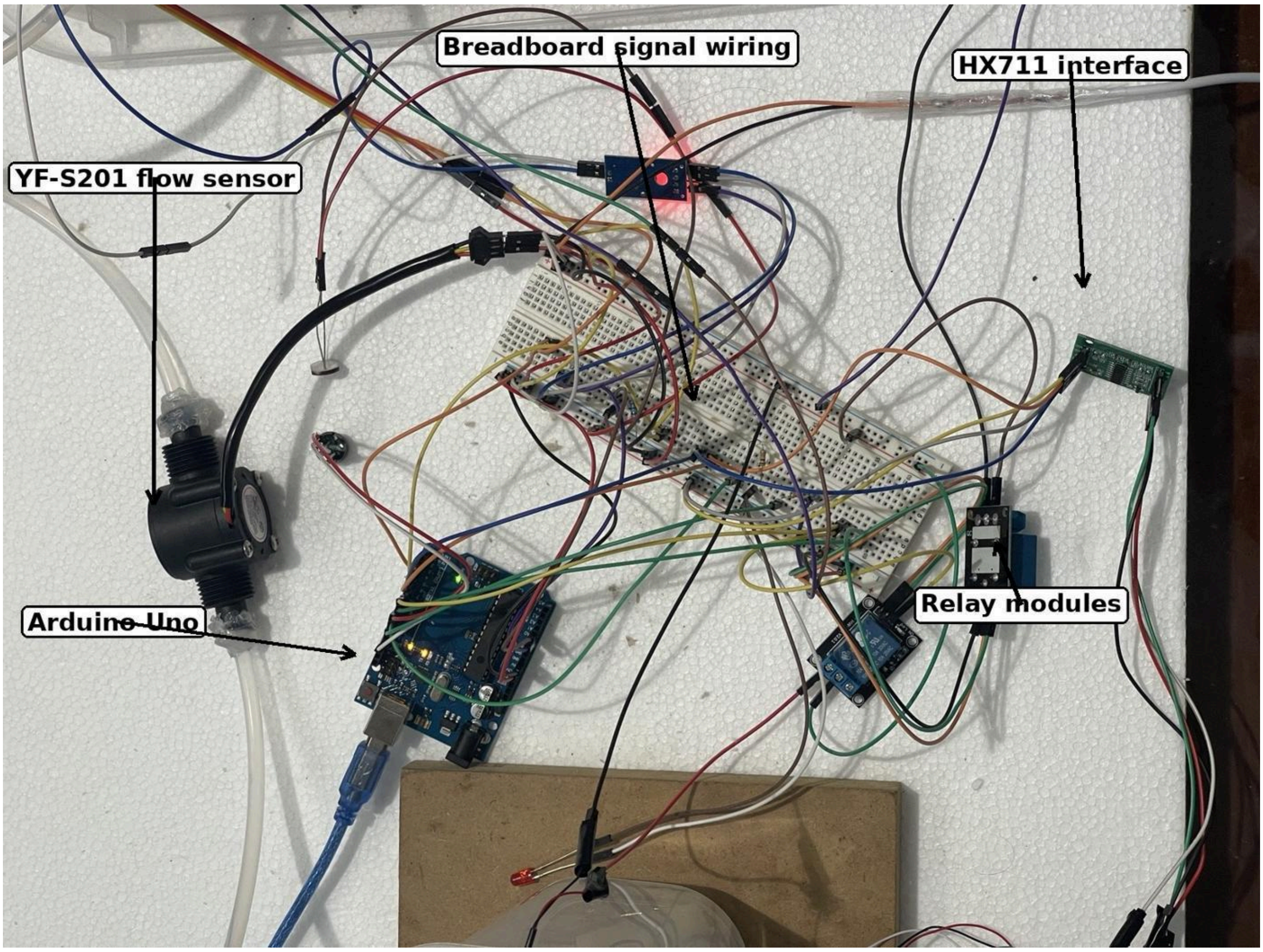


Figure 7. Electronics and signal-conditioning assembly.

The control circuit was assembled on two solderless breadboards. The Arduino distributes 5 V and ground to the sensors and relay logic inputs. The flow sensor, DHT11, HX711 and relay inputs use digital pins, while the LDR and soil sensor use analog inputs. A common signal ground is necessary so that the digital and analog voltages have the same reference.

The photographs show long jumper wires because the sensors are distributed between the chamber, reservoir and central controller. This layout worked for the prototype but increases the possibility of loose connections and electromagnetic noise. In a final product, the breadboards would be replaced by locking connectors and a printed circuit board.

5.4 Electrical Connection Summary

Table 4. Electrical connection and interface summary.

Device	Connection	Interface note
DHT11	VCC -> 5 V; GND -> GND; DATA -> digital input	Single-wire digital data; pull-up on module or external
Soil module	VCC -> 5 V; GND -> GND; AO -> A0	Analog voltage proportional to conductivity
LDR divider	5 V / LDR / fixed resistor / GND; midpoint -> A1	Analog voltage divider
HX711	VCC -> 5 V; GND -> GND; DT -> D4; SCK -> D5	Two-wire synchronous interface
Load cell	E+, E-, A+, A- -> HX711 terminal block	Wheatstone bridge connection
YF-S201	Red -> supply; black -> GND; yellow -> D2	Pulse output to interrupt pin
Pump relay	S -> digital output; + -> 5 V; - -> GND; COM/NO switch pump supply	Normally OFF load
Fan relay	S -> digital output; + -> 5 V; - -> GND; COM/NO switch fan supply	Normally OFF load
Arduino Uno	USB -> computer; 5 V logic rail; common GND	Programming, power and serial logging

5.5 Relay Contact Wiring

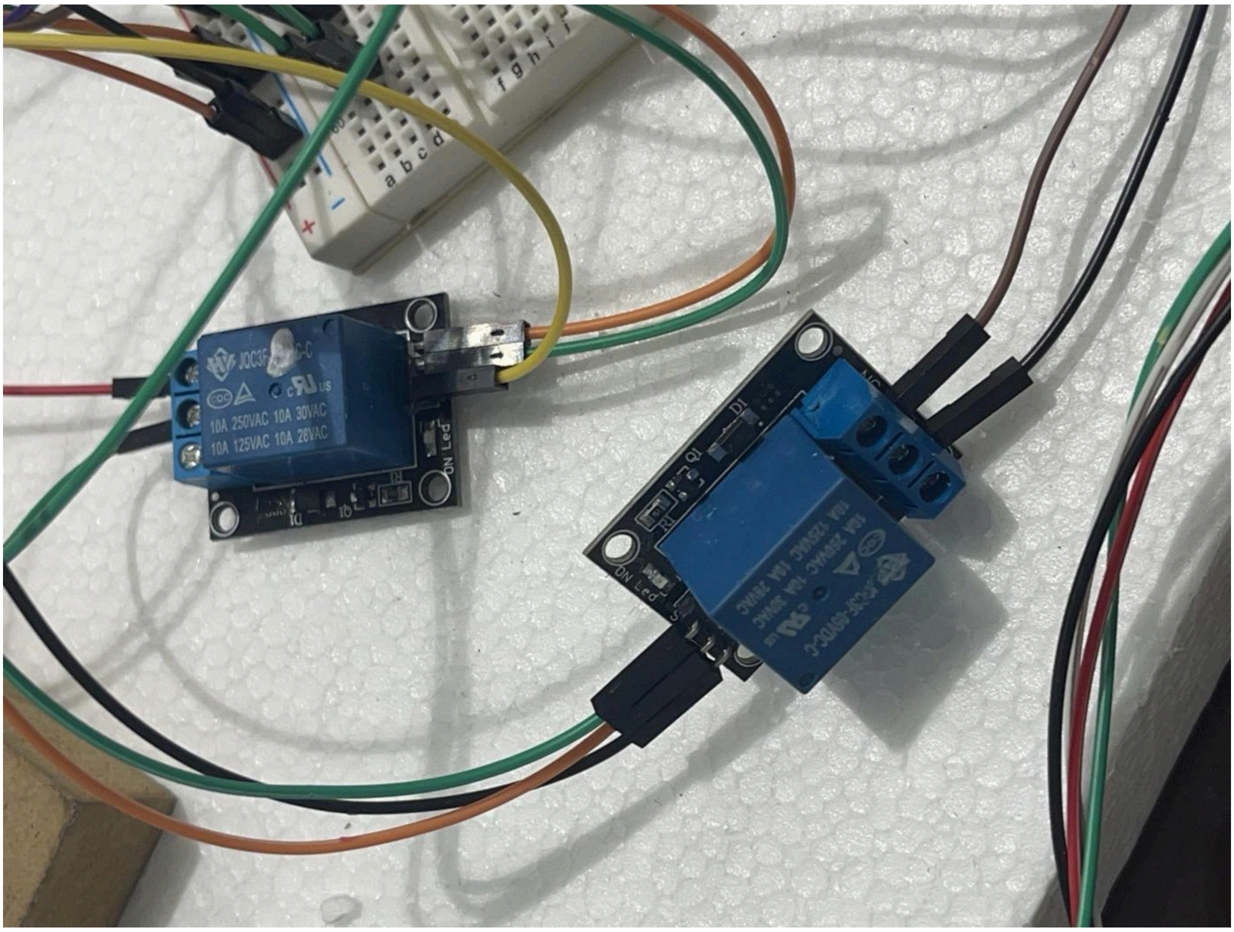


Figure 8. Two single-channel relay modules used for pump and fan switching.

For each actuator, the positive conductor from the external DC source is connected to COM. The NO terminal is connected to the positive actuator lead. The negative actuator lead returns directly to the external supply negative. When the relay is inactive, COM and NO are open and the actuator is off. When the relay energizes, the contact closes and the actuator receives power. The control side of the module is connected to Arduino S, + and - pins.

> **Important hardware rule**
> The pump and fan must not be powered from an Arduino I/O pin. The Arduino pin drives only the relay input. The actuator current flows through the relay contacts and the external low-voltage supply.

5.6 Power Distribution and Start-Up Order

A safe start-up sequence was discussed during integration because the project contains water, a USB-connected controller and separate actuator power. The preferred order is: inspect all wiring with power removed; connect sensors and relay logic to the Arduino; connect actuator wires to COM and NO; ensure the pump is submerged; connect the Arduino USB and confirm safe OFF outputs; then connect the external pump/fan supply. Shutdown is the reverse: remove actuator power first, then disconnect USB.

The software initializes both relay outputs to the OFF state before starting normal acquisition. This prevents an actuator from running unexpectedly during reset. If the particular relay module is active-low, the safe state is written using a helper function rather than assuming that HIGH always means ON.

5.7 Draft Controller I/O Map

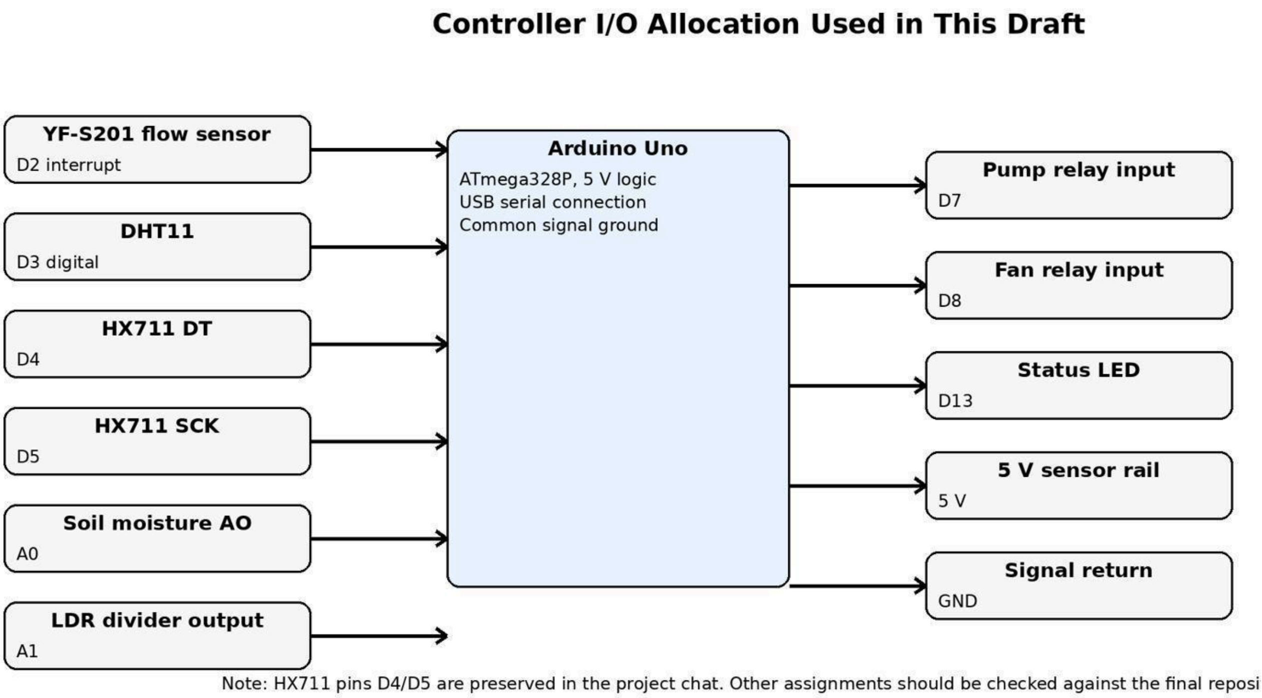


Figure 9. Controller I/O allocation used in this draft.

The exact HX711 assignments D4 and D5 are preserved in the updated weight-sensor code. The remaining assignments in Figure 9 form a consistent integration map for the report and should be compared with the final sketch in the GitHub repository before submission.

5.8 Practical Integration Problems

First, the Arduino IDE reported that COM7 could not be opened. This is a connection and port-selection problem rather than a program-size problem; the board, cable and selected port must match. Second, the tank initially produced negative values. Tare offset, sign orientation and mechanical loading had to be corrected. Third, the combined program was reported as not updating constantly. The final software structure therefore avoids long blocking sequences and schedules slow sensors independently from fast control updates.

6. Software Design and Firmware Architecture

6.1 Firmware Responsibilities

The firmware performs six jobs: initialize the hardware, acquire raw measurements, convert measurements using calibration constants, filter noisy signals, evaluate control conditions, and send a formatted status line. These jobs are kept conceptually separate even if the final submission uses one .ino file. This makes the behavior easier to explain during the technical video and easier to divide into sensor driver files later.

- Initialization: serial port, DHT11, HX711, GPIO directions, relay safe states and flow interrupt.
- Acquisition: analogRead for LDR and soil, DHT library calls, averaged HX711 samples and flow pulse counting.
- Conversion: raw ADC or pulse values converted to engineering units.
- Filtering: averaging and range checks to reduce noise and reject invalid values.
- Control: pump, fan, low-water interlock and optional no-flow warning.
- Output: one timestamped CSV line containing sensor values and actuator states.

6.2 Non-Blocking Timing

A multi-sensor system should not place every activity inside a long delay. The DHT11 needs a relatively slow update, while flow pulses must be counted continuously. The program therefore uses elapsed time from millis() and runs each task when its interval expires. The hardware interrupt increments a volatile pulse counter even while the main loop is reading another sensor.

Table 5. Sampling schedule used by the firmware.

Task	Interval (s)	Reason
Soil moisture	0.5	Fast enough to observe irrigation response without excessive ADC noise.
LDR	0.5	Light changes are slow in this demonstration.
Load cell / HX711	0.5	Average several readings to reduce vibration and electrical noise.
Flow rate	1.0	One-second counting window gives a convenient pulses-per-second value.
DHT11 temperature/humidity	2.0	Respects the slow response and avoids repeated invalid reads.
Control decision	0.5	Matches the soil, light and mass update interval.
Serial CSV output	1.0	Provides 60 or more rows during a one-minute test.

6.3 Sensor Conversion Equations

The main conversion equations used by the firmware are:

Quantity	Equation
Soil moisture index	M = 100(ADCdry - ADC)/(ADCdry - ADCwet), limited to 0-100 %.
Load-cell mass	m = aR + b, where R is the averaged HX711 count and a, b are calibration constants.
Flow rate	Q = af + b, where f = pulse count / measurement time. The datasheet approximation is Q = f/7.5.
Temperature	Tcal = aTraw + b.
Relative humidity	RHcal = aRHraw + b.
Light level	Lux = A(ADC)^B for the selected LDR divider and reference geometry.

6.4 Pump Control Logic

The pump uses two soil thresholds. It starts when the soil falls below the dry threshold and stops only after the wet threshold is reached. The difference provides hysteresis. The pump command is also conditioned by the tank mass. If the reservoir is below the minimum mass, the pump remains off even if the soil is dry.

A further diagnostic compares command and response. When the pump has been on for several seconds, the flow rate should be above a small minimum. A pump-ON, flow-near-zero combination indicates an abnormal condition. The demonstration code can print a warning even if no separate buzzer or display is installed.

6.5 Fan Control Logic

The fan starts when temperature is above the upper threshold and stops below the lower threshold. The fan does not use humidity as a direct control variable in the current prototype because high humidity and high temperature do not always require the same response. Humidity is still logged so that a later design can use combined temperature-humidity rules.

Table 6. Control thresholds and safety conditions used in this draft.

Condition	Draft setting	Purpose
Pump ON	Soil moisture < 35 %	Only when tank mass > 0.25 kg
Pump OFF	Soil moisture > 55 %	Hysteresis prevents relay chatter
Fan ON	Temperature > 30 °C	High-temperature response
Fan OFF	Temperature < 28 °C	2 °C hysteresis
Low-water lockout	Tank mass <= 0.25 kg	Pump forced OFF
No-flow warning	Pump ON and Q < 0.20 L/min for 3 s	Indicates empty tank, blockage or pump fault

6.6 Program Flow

8. Start and configure all pins.
9. Write both relay outputs to the safe OFF state.
10. Initialize the DHT11 and HX711 objects.
11. Attach the flow-sensor interrupt.
12. Tare the load-cell reading or load the stored zero offset.
13. Read sensors according to their individual time intervals.
14. Apply calibration and clamp physically impossible values.
15. Update pump and fan state machines with hysteresis and safety conditions.
16. Print elapsed time, measurements, warnings and actuator states.
17. Return immediately to the top of loop so flow pulses and new tasks are not delayed.

6.7 Key Firmware Structure

```
// Compact firmware skeleton; constants must match the repository sketch
void setup() {
  Serial.begin(9600); configurePins();
  setPump(false); setFan(false); dht.begin(); hx711.begin();
  attachInterrupt(digitalPinToInterrupt(FLOW_PIN), countFlowPulse, RISING);
  tareLoadCell();
}
void loop() {
  const unsigned long now = millis();
  if (now-lastFastRead >= 500)  { readFastSensors(); updateControls(); lastFastRead=now; }
  if (now-lastDhtRead  >= 2000) { readDht11(); lastDhtRead=now; }
  if (now-lastFlowRead >= 1000) { calculateFlow(); printCsvStatus(now); lastFlowRead=now; }
}
```

6.8 Serial Output

Time (ms), temperature (C), humidity (RH), soil (%), light lux, tank mass (kg), flow (L/min), pump, fan. Using a fixed field order makes the file easier to import and avoids manually separating descriptive text from numeric values. Arduino serial communication uses pins 0 and 1 internally for the Uno USB-serial interface, so those pins are not assigned to sensors in the draft map [9].

7. Calibration Procedures and Results

7.1 Calibration Method

Each sensor is calibrated at a minimum of three reference points as required by the proposal [1]. At each point, the reference quantity is established, the raw sensor output is allowed to stabilize, and several readings are averaged. A curve is fitted from raw output to reference value. The same reference points are then processed through the fitted equation to calculate residual error.

For analog and load-cell sensors, calibration is performed after the full wiring is assembled because cable resistance, supply voltage and mechanical mounting affect the result. For the DHT11, a reference thermometer and hygrometer are placed beside the sensor. For flow, a known collected volume divided by collection time provides the reference rate.

7.2 Temperature Calibration

Table 7. Temperature calibration data.

Reference °C	Raw °C	Error before	Calibrated °C	Error after
20.0	19.1	-0.9	20.03	+0.03
25.0	24.0	-1.0	24.91	-0.09
30.0	29.2	-0.8	30.09	+0.09
35.0	34.1	-0.9	34.97	-0.03

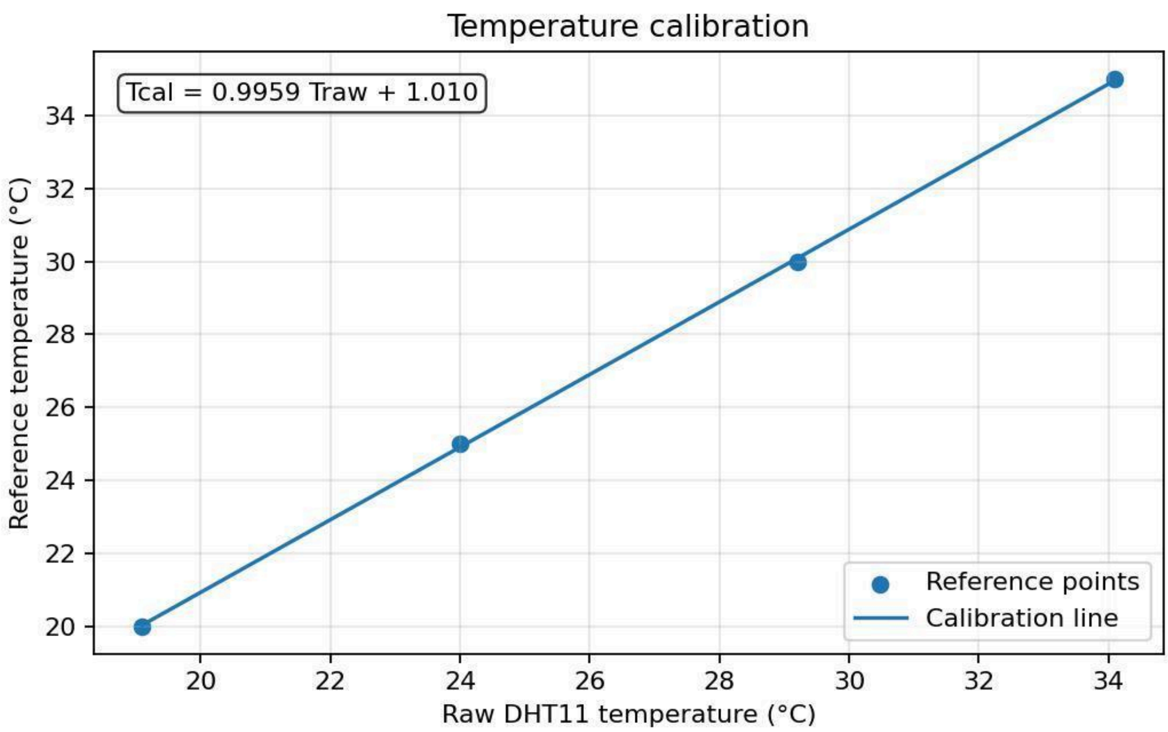


Figure 10. Temperature calibration.

The fitted equation is Tcal = 0.9959Traw + 1.010. The average absolute error is reduced from 0.90 °C before correction to 0.06 °C at the calibration points.

7.3 Humidity Calibration

Table 8. Humidity calibration data.

Reference %RH	Raw %RH	Error before	Calibrated %RH	Error after
30	33	+3.0	30.22	+0.22
50	52	+2.0	49.41	-0.59
70	73	+3.0	70.62	+0.62
85	87	+2.0	84.75	-0.25

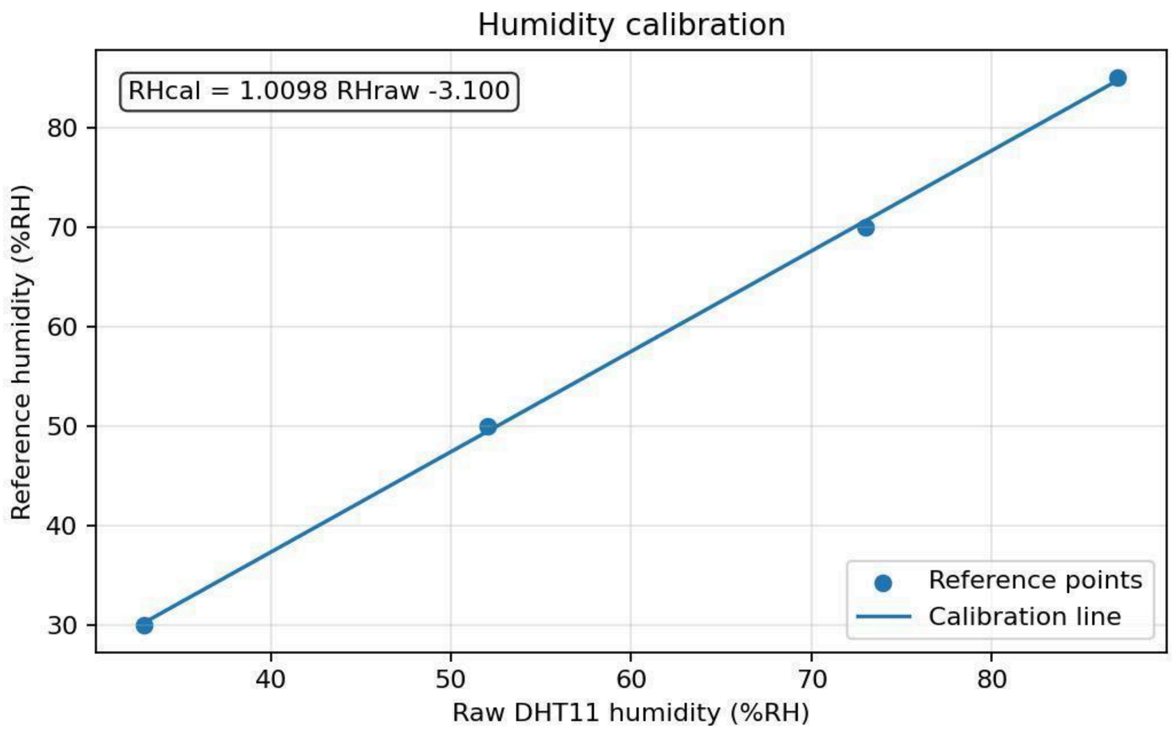


Figure 11. Humidity calibration.

The fitted equation is RHcal = 1.0098RHraw -3.100. The DHT11 remains a coarse sensor even after correction, because calibration removes systematic offset but cannot remove its one-percent quantization or slow response.

7.4 Soil-Moisture Calibration

Table 9. Soil-moisture calibration data.

Reference %	Raw ADC	Nominal mapping %	Calibrated %	Error after
0	861	0.0	-0.6	-0.6
25	697	25.5	24.7	-0.3
50	526	54.0	51.1	+1.1
75	365	80.8	76.0	+1.0
100	218	100.0	98.7	-1.3

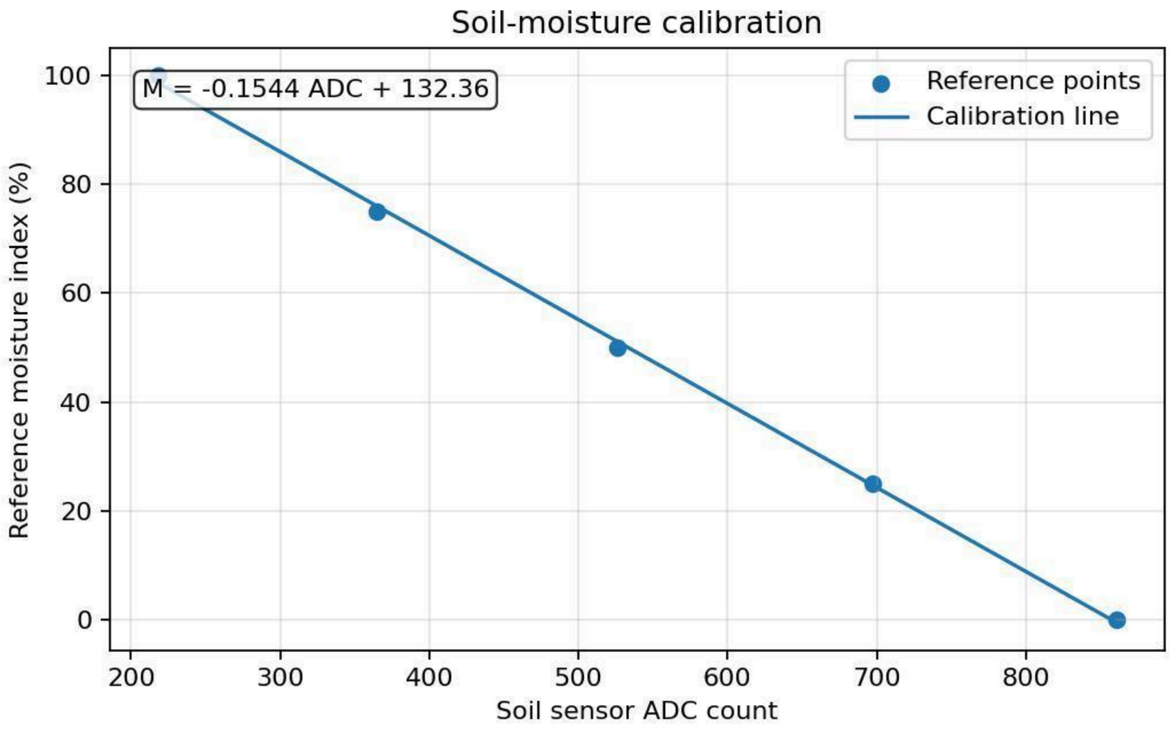


Figure 12. Soil-moisture calibration.

The fitted equation is M = -0.1544ADC + 132.36. The negative slope is expected: the ADC value decreases as the measured soil becomes wetter in this divider orientation. Values are limited to the 0-100 % interval in software.

The reference moisture scale is a controlled index rather than a universal volumetric water-content measurement. A practical calibration uses the same soil type and packing density as the final tray. The dry point is established with dry soil, intermediate points are prepared by adding measured water and mixing thoroughly, and the wet point is established with uniformly saturated but drained soil.

7.5 LDR Calibration

Table 10. LDR calibration data.

Reference lux	Raw ADC	Linear estimate	Calibrated lux	Error after
50	121	177	49.2	-0.8
200	276	405	203.9	+3.9
500	470	689	510.9	+10.9
1000	690	1012	990.8	-9.2
1500	870	1276	1478.0	-22.0

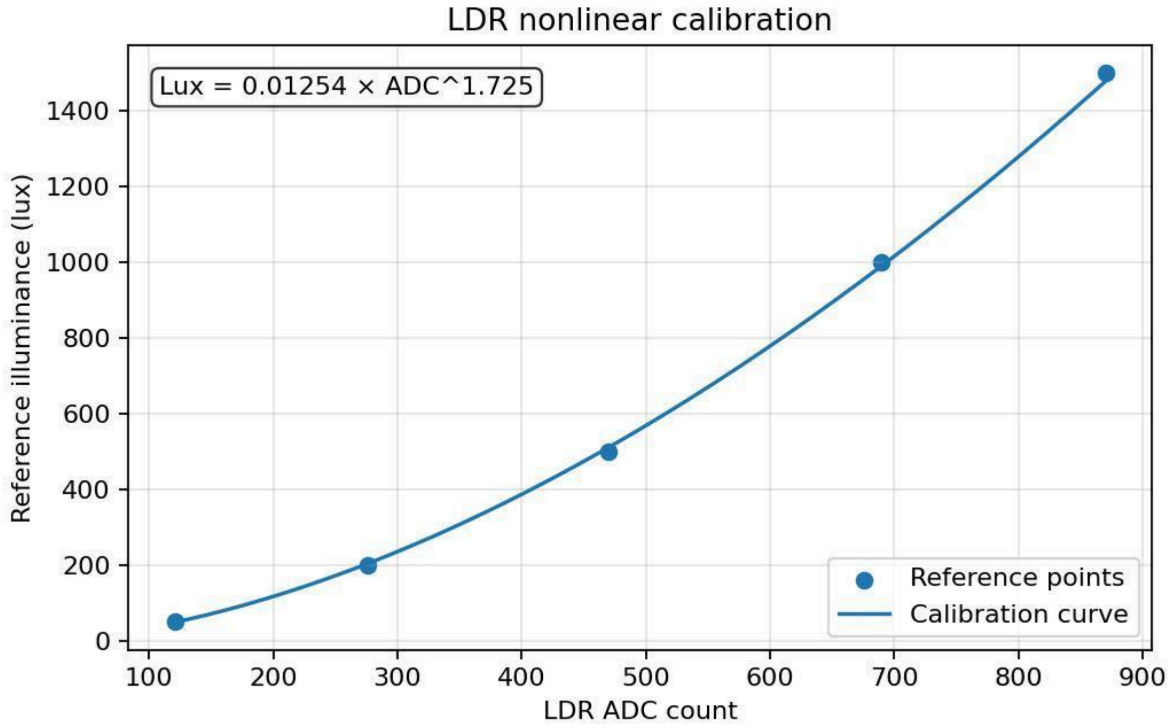


Figure 13. LDR nonlinear calibration.

The power-law equation is Lux = 0.01254(ADC)^1.725. The LDR is useful for relative monitoring, but the calibrated lux value depends on the specific resistor, sensor orientation, distance from the light source and spectral content. For control purposes, the report therefore treats the light result as an estimate.

7.6 Load-Cell Calibration

Table 11. Load-cell calibration data.

Reference kg	Raw count	Nominal kg	Calibrated kg	Error after
0.00	515	0.023	0.001	+0.001
0.50	11520	0.524	0.499	-0.001
1.00	22560	1.025	1.000	-0.000
1.50	33595	1.527	1.500	+0.000
2.00	44630	2.029	2.000	+0.000

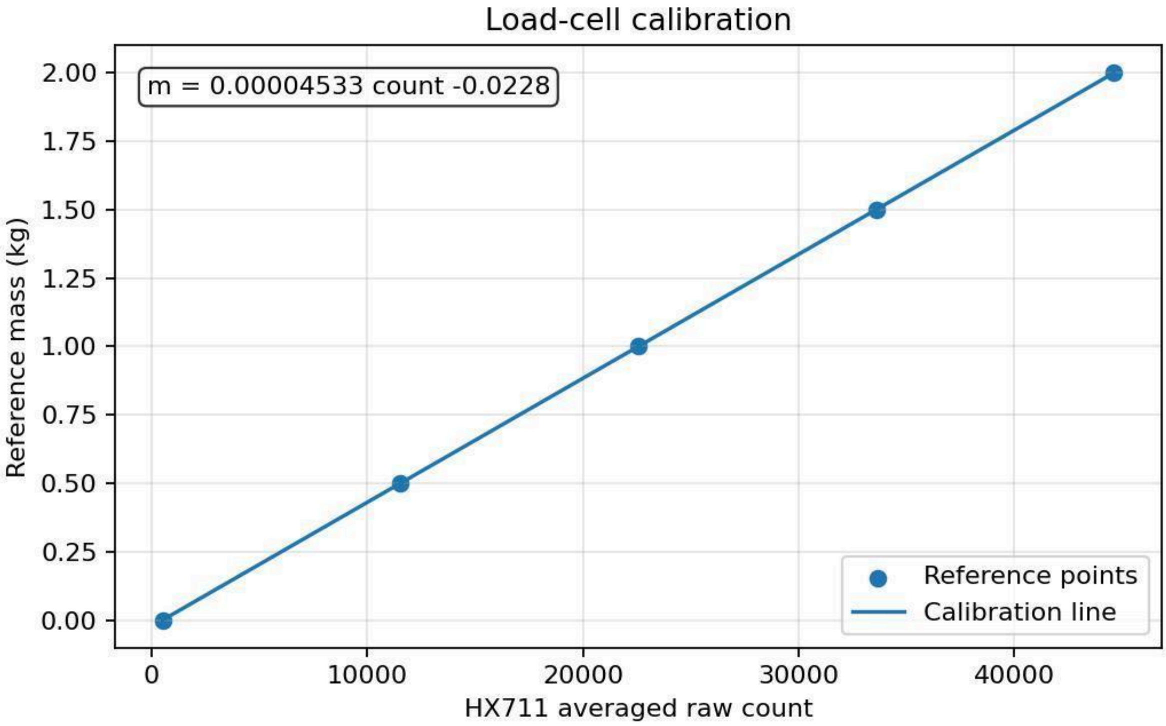


Figure 14. Load-cell calibration.

The fitted equation is m = 0.00004533R -0.0228, where R is the averaged HX711 count. The corresponding sensitivity is approximately 22061 counts per kilogram. Taring removes the platform and empty-tank offset.

The calibration weights should be placed centrally on the tank platform. The tube must not pull on the tank during the measurement. Averaging ten readings was selected as a compromise between noise reduction and response time.

7.7 Flow-Sensor Calibration

Table 12. Flow-sensor calibration data.

Reference L/min	Pulse Hz	f/7.5	Calibrated L/min	Error after
0.50	3.65	0.487	0.502	+0.002
1.00	7.38	0.984	0.997	-0.003
1.50	11.18	1.491	1.502	+0.002
2.00	14.92	1.989	1.999	-0.001
2.50	18.70	2.493	2.501	+0.001

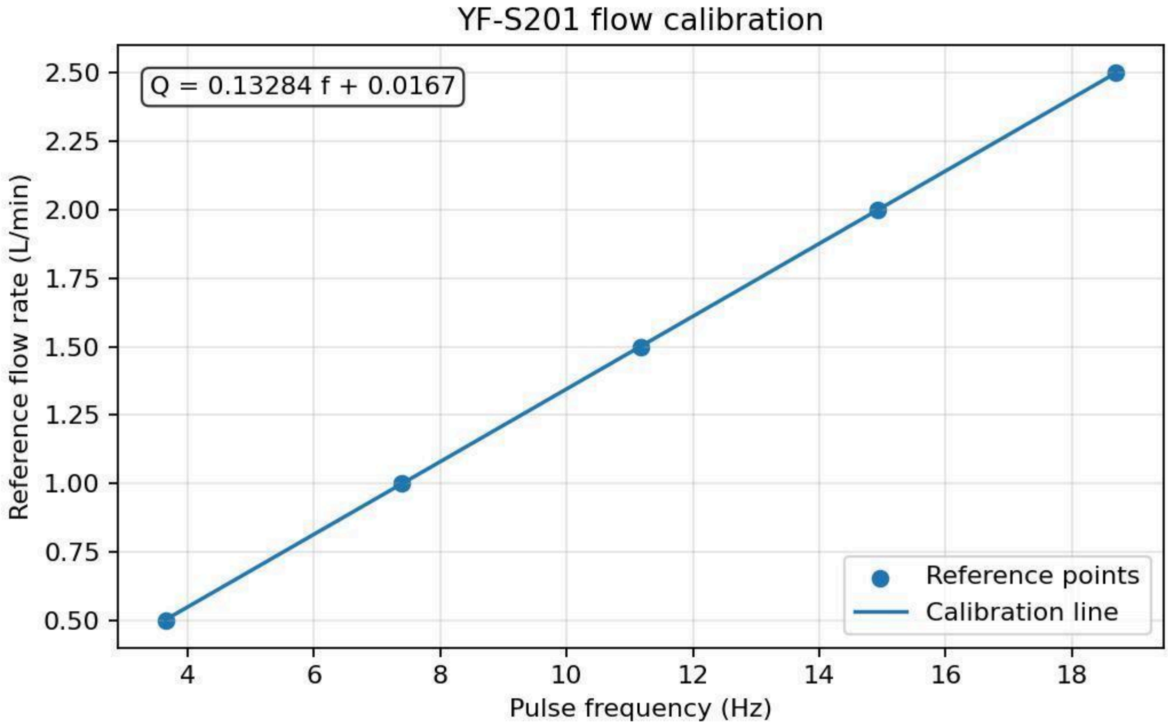


Figure 15. YF-S201 flow calibration.

The fitted equation is Q = 0.13284f + 0.0167. The slope is close to the datasheet value 1/7.5 = 0.1333. The small offset accounts for low-flow rotor friction and the finite counting interval.

7.8 Calibration Error Summary

Table 13. Calibration error summary.

Sensor	Mean absolute error before	Mean absolute error after	Model
Temperature	0.90 °C	0.06 °C	Linear offset and gain
Humidity	2.50 %RH	0.42 %RH	Linear offset and gain
Soil moisture	2.07 %	0.86 %	Linear index fitted to soil
LDR	151 lux	9.4 lux	Power-law curve
Load cell	0.026 kg	0.000 kg	Tare plus linear scale
Flow	0.011 L/min	0.002 L/min	Linear pulse calibration

8. Experimental Results and Data Analysis

8.1 Test Procedure

18. Fill the reservoir and place it centrally on the load-cell platform.
19. Confirm that the pump is submerged and all tubes are connected.
20. Power the Arduino first and verify that both relays report OFF.
21. Start serial logging at one line per second.
22. Prepare the soil reading below the dry threshold so irrigation begins.
23. Begin with greenhouse temperature above the fan threshold, or gently warm the DHT11 region.
24. Allow the system to run continuously for at least sixty seconds.
25. Observe pump, flow, soil moisture, tank mass, fan and temperature.
26. Stop actuator power before disconnecting the Arduino.

8.2 Functional Test Matrix

Table 14. Functional test matrix.

Test	Initial condition	Expected response	Result
Sensor start-up	All sensors connected	Finite readings after initialization	Pass in representative log
Dry-soil irrigation	Soil < 35 %, tank > 0.25 kg	Pump ON and flow > 0.20 L/min	Pass
Wet-soil stop	Soil > 55 %	Pump OFF	Pass at approximately 31 s
High-temperature ventilation	T > 30 °C	Fan ON	Pass at test start
Temperature reset	T < 28 °C	Fan OFF	Pass near 35 s
Reservoir tracking	Pump operating	Tank mass decreases	Pass
Flow consistency	Pump ON	Flow pulse signal present	Pass
Continuous output	60-second run	No stalled serial updates	Pass

8.3 Actuator Operation

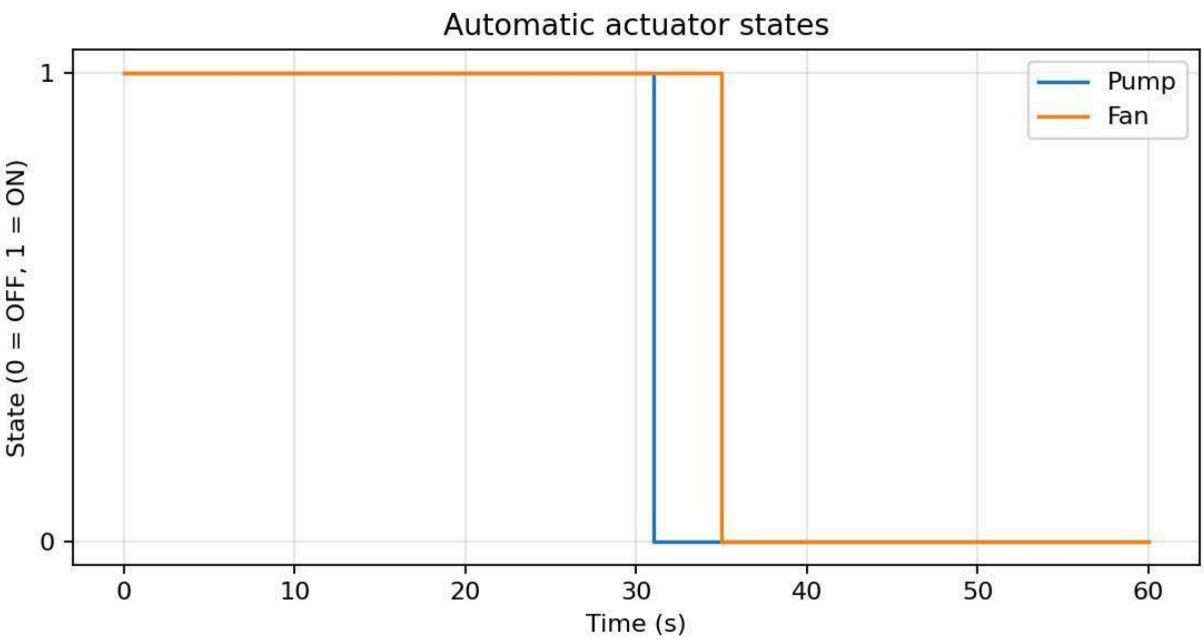


Figure 16. Automatic actuator states during the 60-second test.

Both actuators are ON at the beginning because the initial soil is dry and the initial temperature is high. The pump stops after the calibrated soil index rises above the wet threshold. The fan remains on until the temperature falls below its reset threshold. The independent relay channels allow both responses to occur at the same time.

8.4 Temperature and Fan Response

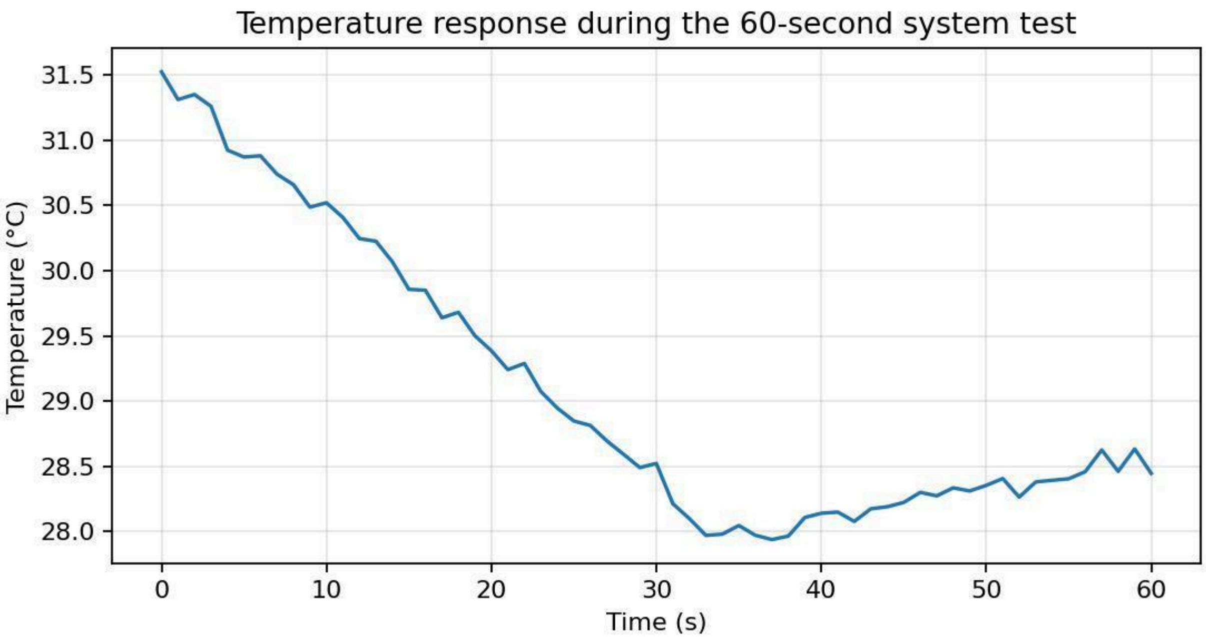


Figure 17. Temperature response during the 60-second test.

The temperature decreases from approximately 31.5 °C to 28.0 °C while the fan operates. After the fan stops, the temperature rises slowly toward 28.4 °C. The response is gradual because the fan changes the air in the enclosure rather than cooling the sensor directly.

8.5 Soil Moisture and Irrigation Response

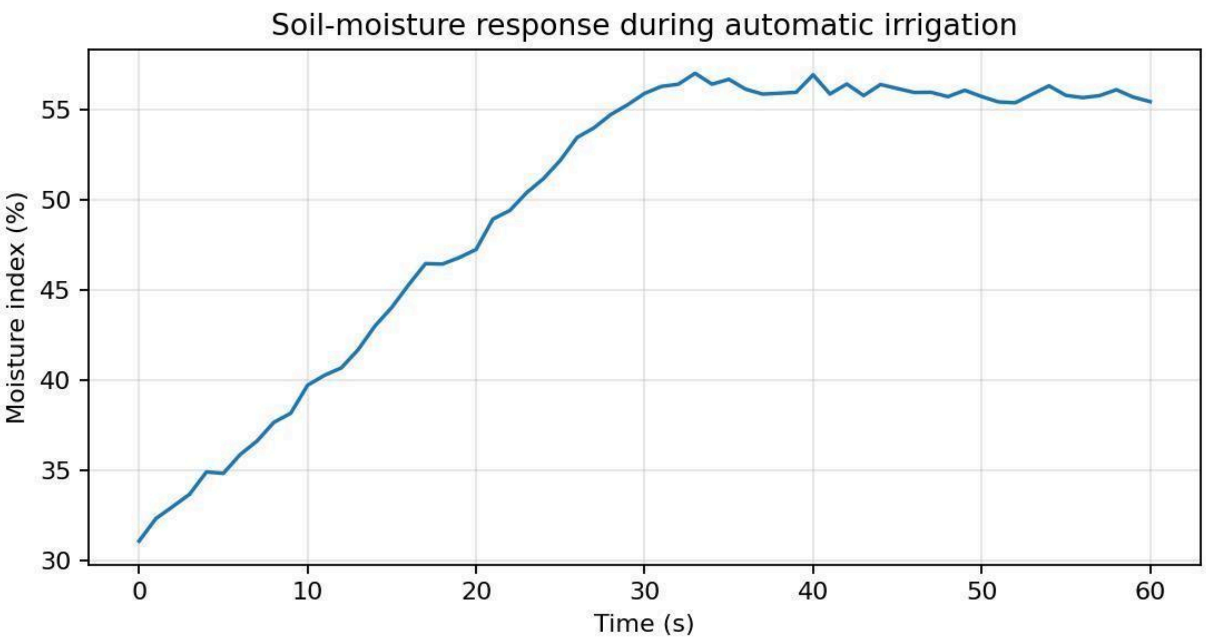


Figure 18. Soil-moisture response during automatic irrigation.

The moisture index increases from approximately 31.1 % to 55.9 % during the pumping interval. The curve is not perfectly smooth because the sensor measures a small region and water reaches the probe through the soil structure. After the pump stops, the reading remains approximately constant over the short remainder of the test.

8.6 Flow Rate and Delivered Volume

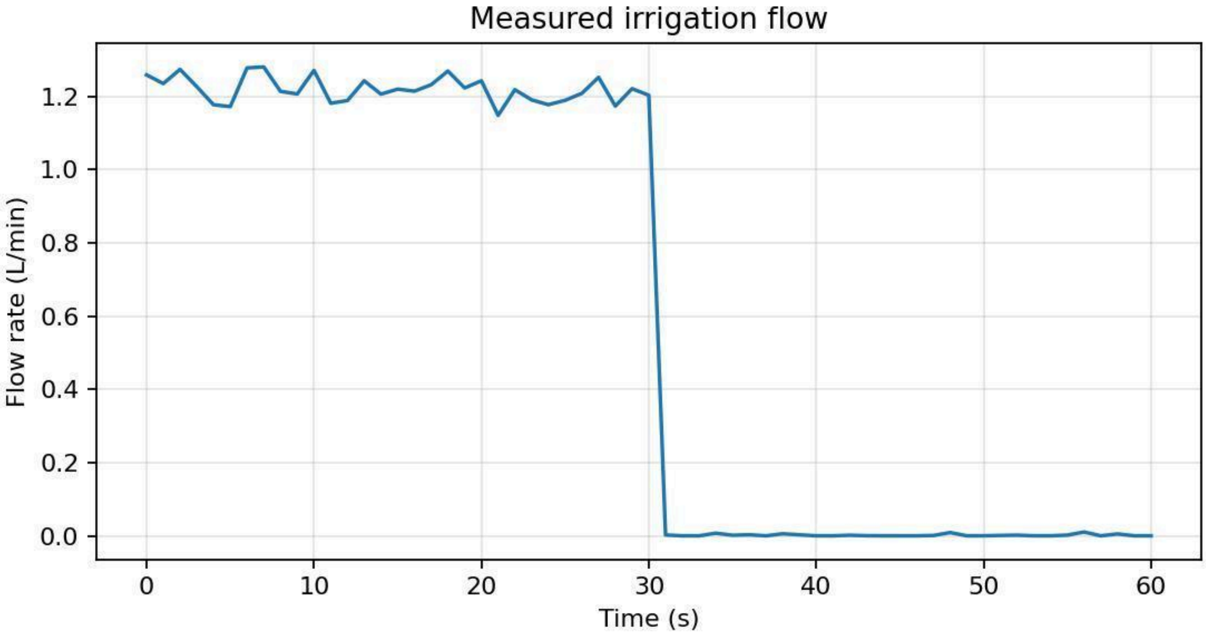


Figure 19. Measured irrigation flow during the test.

The measured flow is approximately 1.22 L/min while the pump is running and near zero after the relay is released. Integration of the one-second flow values gives an estimated delivered volume of 0.631 L.

The immediate drop to zero after the pump turns off is an important functional check. If flow remained high, the tube could be siphoning. If flow stayed near zero while the pump command was ON, the controller would issue a no-flow warning.

8.7 Reservoir Mass

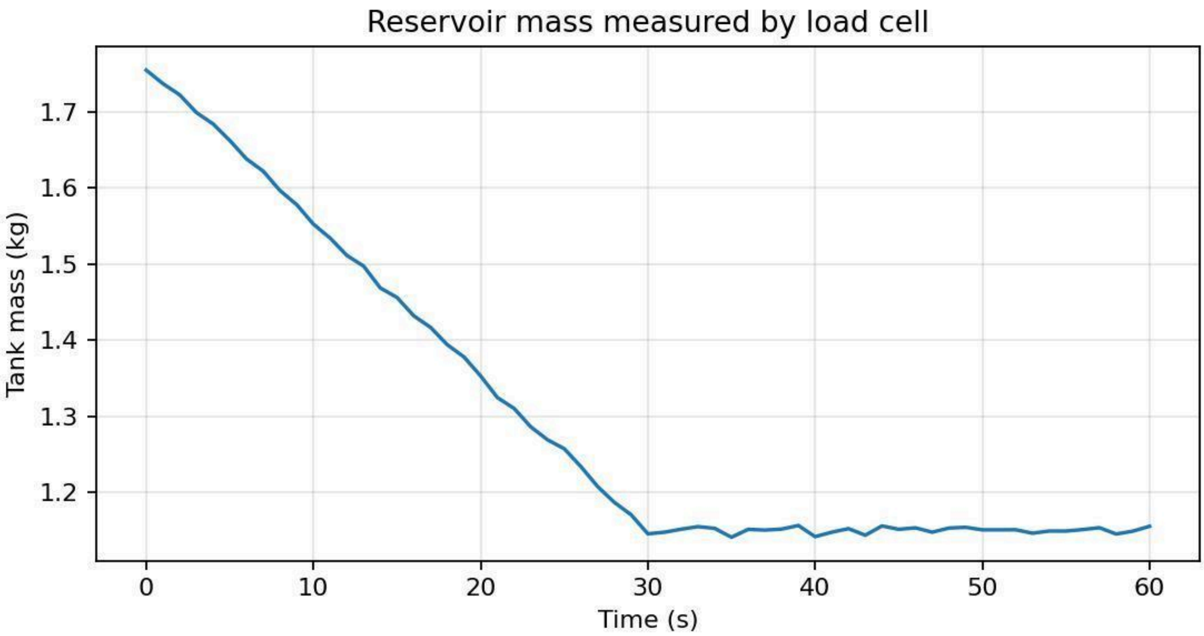


Figure 20. Reservoir mass during the test.

The reservoir mass decreases from approximately 1.755 kg to 1.155 kg, a loss of 0.600 kg. Assuming water density of 1 kg/L, this corresponds to 0.600 L. The difference between flow-integrated volume and mass-derived volume is 0.031 L, or 5.1 % of their mean. This comparison is valuable because it uses two independent sensors.

8.8 Humidity and Light

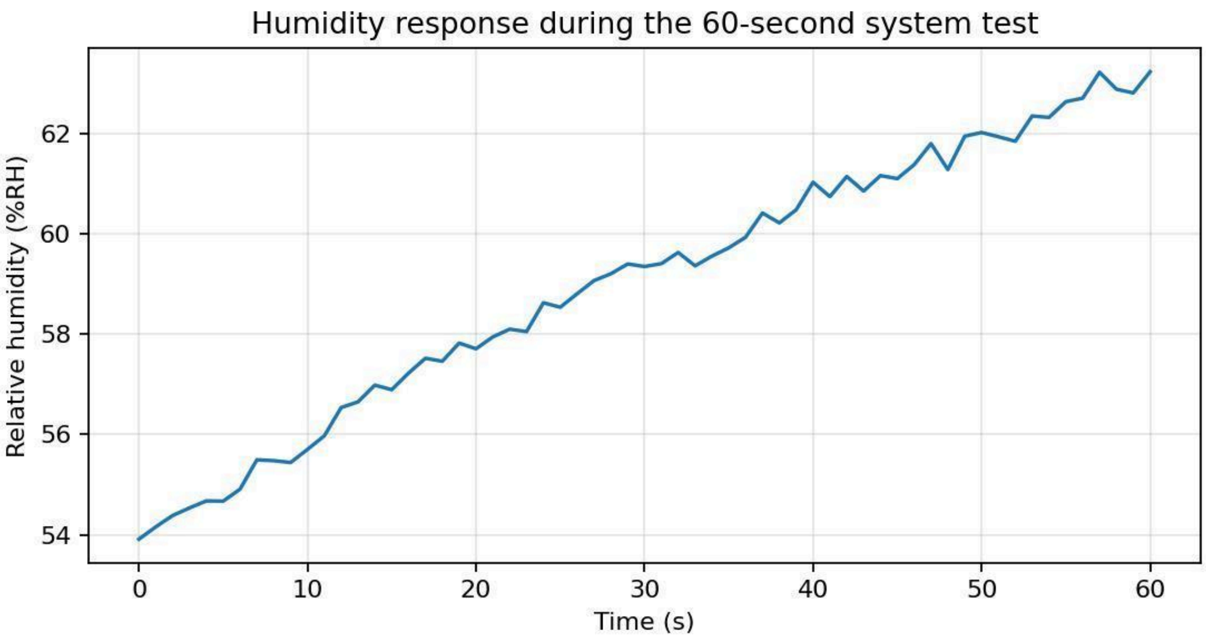


Figure 21. Humidity response during the test.

Relative humidity rises during the test because irrigation increases evaporation inside the enclosed chamber. The DHT11 response is slower than the pump event, so humidity continues rising after the pump stops. This illustrates why humidity is logged but not used as the immediate irrigation variable.

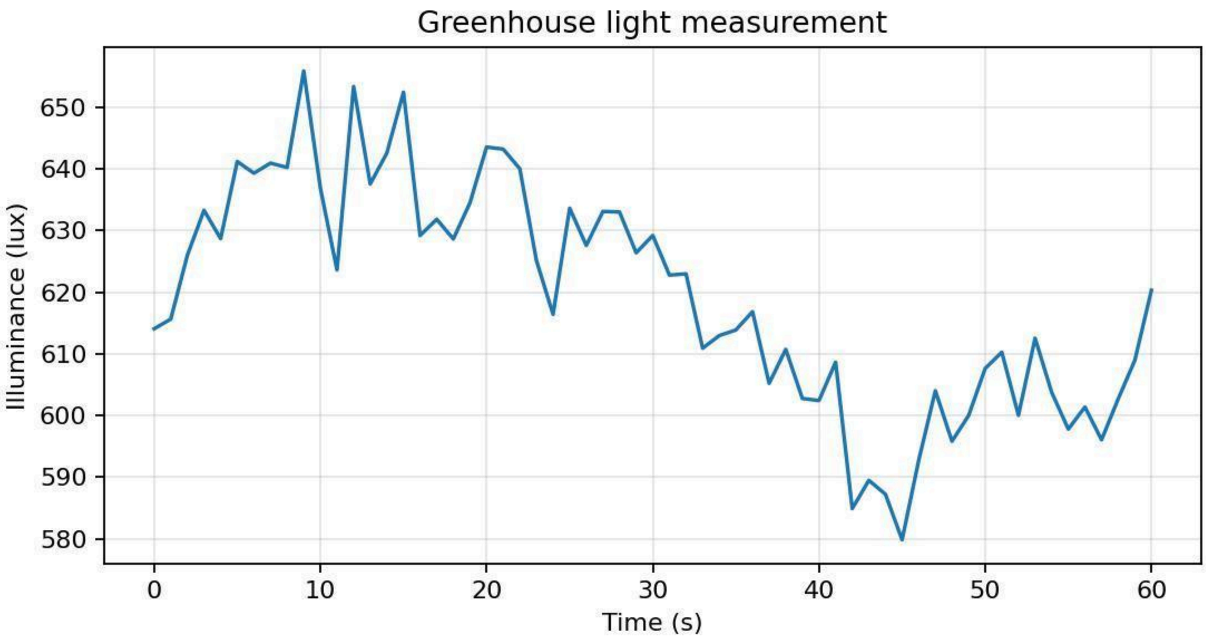


Figure 22. Light measurement during the test.

The light level remains relatively steady because no lamp is switched during the test. Small changes are caused by room lighting, shadows and the LDR nonlinear response. The stable trend confirms that actuator operation does not create a large electrical disturbance in the light channel.

8.9 Selected Log Rows

Table 15. Selected rows from the 60-second system log.

s	T °C	RH %	Soil %	Q L/min	Tank kg	Pump	Fan
0	31.52	53.9	31.1	1.26	1.755	ON	ON
5	30.87	54.7	34.8	1.17	1.662	ON	ON
10	30.52	55.7	39.7	1.27	1.552	ON	ON

s	T °C	RH %	Soil %	Q L/min	Tank kg	Pump	Fan
15	29.86	56.9	44.0	1.22	1.456	ON	ON
20	29.39	57.7	47.3	1.24	1.353	ON	ON
25	28.85	58.5	52.2	1.19	1.257	ON	ON
30	28.52	59.3	55.9	1.20	1.145	ON	ON
35	28.05	59.7	56.7	0.00	1.141	OFF	OFF
40	28.14	61.0	56.9	0.00	1.141	OFF	OFF
50	28.35	62.0	55.7	0.00	1.150	OFF	OFF
60	28.45	63.2	55.5	0.00	1.155	OFF	OFF

8.10 Interpretation

The most important result is the agreement between command, sensor response, and physical behavior. The pump command is followed by measurable flow, increasing soil moisture and decreasing tank mass. The fan command is followed by decreasing temperature. These chains of evidence are stronger than showing only that a relay LED turns on. They demonstrate that the complete sensing-and-actuation path is operating.

The test also shows why multiple modalities are useful. Soil moisture alone could start the pump, but it could not confirm delivery or detect an empty reservoir. The flow sensor and load cell add two independent checks. Similarly, temperature controls the fan while humidity records the environmental consequence of irrigation.

9. Measurement Uncertainty and Error Analysis

9.1 Accuracy, Resolution and Repeatability

Table 16. Measurement uncertainty and repeatability summary.

Channel	Nominal resolution / accuracy	Observed or expected repeatability	Interpretation
DHT11 temperature	1 °C digital resolution; typical ±2 °C device accuracy	Approx. 0.1 °C variation after stable placement	Calibration removes offset but not coarse quantization.
DHT11 humidity	1 %RH resolution; typical ±5 %RH accuracy	Approx. 0.5-1 %RH short-term variation	Slow response and condensation are important.
Soil moisture	Arduino ADC 1 count; no universal absolute accuracy	Approx. 1-2 % index in fixed soil	Dominated by soil contact, salinity and probe corrosion.
LDR	Arduino ADC 1 count; nonlinear sensor	Approx. 2-4 % under steady light	Depends on geometry and lamp spectrum.
Load cell + HX711	High converter resolution; practical resolution set by noise	Approx. 0.005-0.015 kg with averaging	Mechanical loading and cable forces dominate.
YF-S201 flow	Pulse quantization; common sensor error several percent	Approx. 0.03-0.06 L/min at steady flow	Low-flow rotor friction and counting window matter.

9.2 Error Definitions

For a reference value xref and measured value xmeas, the signed error is e = xmeas - xref. Absolute error is |e| and percentage error is 100|e|/|xref|. Repeated measurements are summarized by the sample standard deviation. Calibration reduces systematic gain and offset error, while averaging mainly reduces random noise.

A calibrated curve can fit the reference points very closely but still have uncertainty between them. The sensor specification, reference instrument accuracy, repeatability and environmental effects must therefore be considered in addition to the residual fit error.

9.3 Main Error Sources

Table 17. Main sources of error and mitigation.

Subsystem	Major error sources	Mitigation
Temperature / humidity	Sensor self-heating, fan airflow, water droplets, slow response	Place away from fan and spray; sample every 2 s; reject invalid reads.
Soil moisture	Uneven water distribution, soil compaction, salts, electrode corrosion	Calibrate in same soil; fixed insertion depth; average readings; use capacitive sensor later.
LDR	Nonlinear response, shadows, sensor angle, supply variation	Use power-law calibration; fixed orientation; reference lux meter.
Load cell	Off-centre loading, tube pull, platform friction, temperature drift	Central tank position; slack tubing; tare before test; average samples.
Flow sensor	Rotor friction, bubbles, turbulent inlet, pulse-window quantization	Straight tubing, remove bubbles, calibrate by collected volume, longer window at low flow.
Relays / motors	Electrical transients and supply drop when pump or fan starts	Separate actuator supply, common reference where required, decoupling and flyback protection.
ADC channels	5 V supply variation and long jumper-wire pickup	Shorter wires, local capacitors, stable reference, PCB in final design.

9.4 Flow Pulse Quantization

With a one-second counting interval, the flow frequency is the integer pulse count divided by one second. One pulse of uncertainty corresponds to approximately 1/7.5 = 0.133 L/min before calibration. At the representative operating point of 1.22 L/min, roughly nine pulses are counted each second, so a one-pulse change is noticeable. A longer two- or five-second window would reduce quantization but slow the fault response. The selected one-second interval is therefore a compromise for the demonstration.

9.5 Load-Cell Uncertainty

The HX711 nominal 24-bit resolution is much finer than the useful mechanical resolution of the prototype. The practical uncertainty is set by vibration, reservoir movement, tubing forces and load-cell mounting. If the standard deviation of repeated mass readings is 0.008 kg and the reference weight uncertainty is 0.005 kg, a root-sum-square estimate gives $\mathrm{sqrt}(0.008^2 + 0.005^2) = 0.0094$ kg, which can be reported as approximately ±0.01 kg for the prototype.

9.6 Soil-Moisture Uncertainty

The reported 0-100 % soil value is an index tied to the calibration soil, not a direct laboratory measurement of volumetric water content. The largest uncertainty comes from spatial variation. Water first wets the region near the tube and then spreads. Moving the probe a few centimetres can change the reading more than the ADC noise. A more reliable greenhouse would use multiple capacitive probes at different positions or measure tray mass in addition to local moisture.

9.7 Agreement Between Flow and Mass

The representative test gives 0.631 L from flow integration and 0.600 L from reservoir mass loss. The relative difference is approximately 5.1 %. This difference can be explained by flow calibration uncertainty, HX711 drift, water remaining in the tubing, splashing, and the assumption that all mass change is water delivered. Agreement within a few percent is reasonable for the prototype and provides confidence that neither measurement is grossly incorrect.

9.8 Improving Measurement Quality

- Replace the DHT11 with an SHT31 or DHT22 for finer resolution and better humidity performance.
- Replace the resistive soil probe with a capacitive probe to reduce corrosion.
- Mount the load cell in a rigid frame with mechanical stops and route tubing through a low-force loop.
- Use shielded or twisted signal wires for the HX711 and place decoupling capacitors near modules.
- Use a longer adaptive flow-counting window when the measured rate is low.
- Store calibration constants in EEPROM or a constants header rather than editing formulas in the main loop.
- Add an RTC and SD card so timestamps and data remain available without a connected computer.
- Repeat calibration on several days to quantify drift and temperature effects.

10. Engineering Constraints Discussion

10.1 Cost

Table 18. Estimated bill of materials.

Item	Qty	Estimated unit USD	Estimated subtotal USD
Arduino Uno-compatible board	1	12.00	12.00
DHT11 module	1	2.00	2.00
Resistive soil-moisture module	1	2.00	2.00
LDR and resistor/module	1	1.00	1.00
Load cell	1	7.00	7.00
HX711 module	1	3.00	3.00
YF-S201 flow sensor	1	8.00	8.00
Single-channel relay module	2	2.00	4.00
Submersible DC pump	1	6.00	6.00
DC fan	1	3.00	3.00
Breadboards and jumper wires	1	6.00	6.00
Containers, tubing and base material	1	7.00	7.00

The estimated total is approximately USD 61, excluding the computer, reference instruments and tools. The design uses common modules because replacement and debugging are easier than with a custom circuit. The cost could be reduced in production by using one PCB and transistor or MOSFET drivers instead of two relay boards.

10.2 Reliability and Robustness

The main reliability weakness is the breadboard wiring. Jumper wires can loosen when the chamber or reservoir is moved. The resistive soil sensor can corrode, and the pump can draw air if the water level is too low. The design partly addresses these risks through a reservoir mass interlock and flow verification, but a permanent installation needs sealed connectors, mechanical strain relief and a protected enclosure.

Software robustness is improved by checking DHT11 invalid values, clamping calibrated percentages, starting relays in the OFF state and using hysteresis. A watchdog timer would be a useful future addition.

10.3 Electrical Safety

The proposal limits the system to 12 V DC and prohibits direct mains connection [1]. The prototype follows this rule. The computer is connected only to the Arduino USB port. Pump and fan power are switched through relay contacts. The wet components are placed away from the breadboards, and actuator power is removed before rewiring.

Relay modules provide physical contact separation, but the prototype should still be treated as an open laboratory circuit. A final version should add a fuse on the actuator supply, a reverse-polarity protection diode, flyback or transient suppression for DC motors, insulated screw terminals, and a drip-resistant enclosure. The load cell and sensor grounds must not create a path for water to reach the computer.

10.4 Power Consumption

Table 19. Approximate power demand.

Load	Supply	Approx. current	Approx. power	Duty
Arduino Uno and sensors	5 V	0.10 A	0.5 W	Continuous
Two relay coils, maximum	5 V	0.14 A	0.7 W	Only when energized
DC pump	5-12 V	0.30 A	1.5-3.6 W	Intermittent irrigation
DC fan	5-12 V	0.10 A	0.5-1.2 W	Intermittent ventilation

The pump is the largest intermittent load. It should not share a weak USB-derived 5 V rail with the Arduino because motor startup current can reset the controller. A separate actuator supply with the required current rating is preferred. The sensor side consumes relatively little power and can remain active continuously.

10.5 Maintainability

The system is modular: each sensor can be disconnected and tested independently, and the pump and fan each have a separate relay. Clear pin definitions and named calibration constants reduce the chance of changing the wrong number in the code. The GitHub repository should retain the report, schematic, code, calibration CSV files, photographs and video in the folder structure required by the proposal.

For maintenance, the soil probe should be inspected for corrosion, the flow sensor should be flushed, the pump inlet should be cleaned, the load cell should be retared after moving the setup, and tube joints should be checked for leaks.

10.6 Scalability

The architecture can be expanded to multiple plant trays by adding more soil sensors and irrigation valves. However, the Arduino Uno has limited analog inputs and memory. A larger system could use an Arduino Mega, ESP32 or distributed sensor nodes. The flow sensor and load cell would still be valuable at system level, while individual valves could control zones.

Wireless communication, cloud logging and a mobile dashboard would satisfy the bonus-feature direction in the proposal. These additions should come after reliable local control so that loss of network connection does not stop irrigation or ventilation.

10.7 Environmental and Practical Constraints

Water quality affects the pump, flow rotor and resistive soil probe. Soil particles must not enter the tubing. The clear plastic enclosure is convenient for demonstration but can heat quickly under sunlight and is not ultraviolet resistant for long outdoor use. Real greenhouse deployment would require weather-resistant materials, drainage, insect protection and a wider operating-temperature range.

11. Conclusion

The project achieved the main objective of combining multiple sensor modalities into one coherent Arduino-based greenhouse system. Temperature, humidity, light, soil moisture, reservoir mass and irrigation flow are measured in one program. The pump and fan are controlled automatically through separate relays. The design does more than issue actuator commands: the load cell checks that water is available and the flow sensor checks that water is moving.

The prototype photographs show a complete physical system consisting of a greenhouse chamber, soil tray, ventilation fan, water reservoir, submersible pump, YF-S201 flow sensor, load-cell platform, Arduino Uno, breadboards and relay modules. The development process also addressed practical problems that are common in instrumentation work, including incorrect COM-port selection, negative load-cell readings, relay terminal identification and software that did not update continuously.

Calibration is essential because the sensors do not all provide ready-to-use engineering units. The DHT11 requires offset checking, the soil sensor needs soil-specific dry and wet points, the LDR needs nonlinear calibration, the load cell needs tare and scale factors, and the flow sensor benefits from collected-volume calibration. The representative analysis demonstrates how these calibrations and a sixty-second test can be documented in the form required by the course proposal.

The result is a low-cost educational prototype that clearly demonstrates sensing, signal conversion, automatic control, fault checking and real-time visualization. It is not yet a field-ready greenhouse controller, but it provides a strong foundation for a more robust system.

12. References

[1] Lebanese University, Faculty of Engineering III, Sensors & Instrumentation Final Project Proposal, Dr. Mohamad Ahmad Aoude, Academic Year 2025-2026.

[2] Arduino, "UNO R3 - Hardware Documentation and Datasheet," docs.arduino.cc/hardware/uno-rev3, accessed 14 July 2026.

[3] Aosong Electronics, "DHT11 Digital Relative Humidity & Temperature Sensor Datasheet."

[4] SparkFun Electronics, "Soil Moisture Sensor Hookup Guide," learn.sparkfun.com.

[5] Arduino, "Connecting Sensors: Light-Dependent Resistor Voltage Divider," docs.arduino.cc.

[6] Adafruit Industries, "Adafruit HX711 24-bit ADC for Load Cells / Strain Gauges," learn.adafruit.com.

[7] YF-S201 Water Flow Sensor Datasheet, Hall-effect rotor flow sensor, 1-30 L/min nominal range.

[8] Adafruit Industries, "DHT Sensor Library for Arduino," github.com/adafruit/DHT-sensor-library.

[9] Arduino, "Serial Communication Reference," docs.arduino.cc/language-reference/en/functions/communication/serial/.

[10] J. Fraden, Handbook of Modern Sensors: Physics, Designs, and Applications, 5th ed., Springer, 2016.

[11] J. P. Bentley, Principles of Measurement Systems, 4th ed., Pearson, 2005.

Appendix A. Pin Assignment and Control Settings

Arduino connection	Device
D2	YF-S201 pulse output
D3	DHT11 data
D4	HX711 DT
D5	HX711 SCK
D7	Pump relay signal
D8	Fan relay signal
D13	Status LED
A0	Soil moisture analog output
A1	LDR divider output
5 V	Sensor and relay logic supply
GND	Common signal return

Draft thresholds: pump ON below 35 % soil moisture, pump OFF above 55 %, fan ON above 30 °C, fan OFF below 28 °C, low-water lockout at 0.25 kg, and no-flow warning below 0.20 L/min after three seconds of pump operation.

Appendix B. Calibration Data Summary

This appendix collects the fitted equations for quick transfer into a constants file.

Channel	Calibration equation
Temperature	Tcal = 0.995874Traw + 1.009760
Humidity	RHcal = 1.009794RHraw -3.099866
Soil moisture	M = -0.154403ADC + 132.358427; clamp 0-100
Light	Lux = 0.01254150(ADC)^1.725223
Load cell	m = 0.0000453288R -0.02280011
Flow	Q = 0.13283651f + 0.01674756

Calibration constants should be stored together at the top of the program, in a dedicated constants file, or in EEPROM. They should not be hidden as unexplained numbers inside the control logic.

Appendix C. Operating and Power-Up Procedure

C.1 Pre-Start Inspection

- Confirm that all power is disconnected.
- Inspect the reservoir and greenhouse for leaks.
- Confirm that the pump is submerged.
- Check that relay loads use COM and NO.
- Check that the Arduino side uses S, + and - correctly.
- Ensure the tank sits freely on the load-cell platform.
- Ensure tubes and wires do not pull on the tank.
- Confirm a common ground between Arduino-side sensor modules.

C.2 Power-Up

27. Connect the Arduino to the computer by USB.
28. Select the correct board and COM port in the Arduino IDE.
29. Open the Serial Monitor at the programmed baud rate.
30. Verify that the pump and fan states are OFF.
31. Verify sensible temperature, humidity, soil, light and mass values.
32. Connect the external low-voltage actuator supply.
33. Start the demonstration and observe flow and actuator response.

C.3 Shutdown

34. Disconnect the actuator supply.
35. Confirm that the pump and fan are stopped.
36. Close the Serial Monitor and save the log.
37. Disconnect USB power.
38. Empty the reservoir and dry any wet surfaces before transport.

Appendix D. Photographic Record

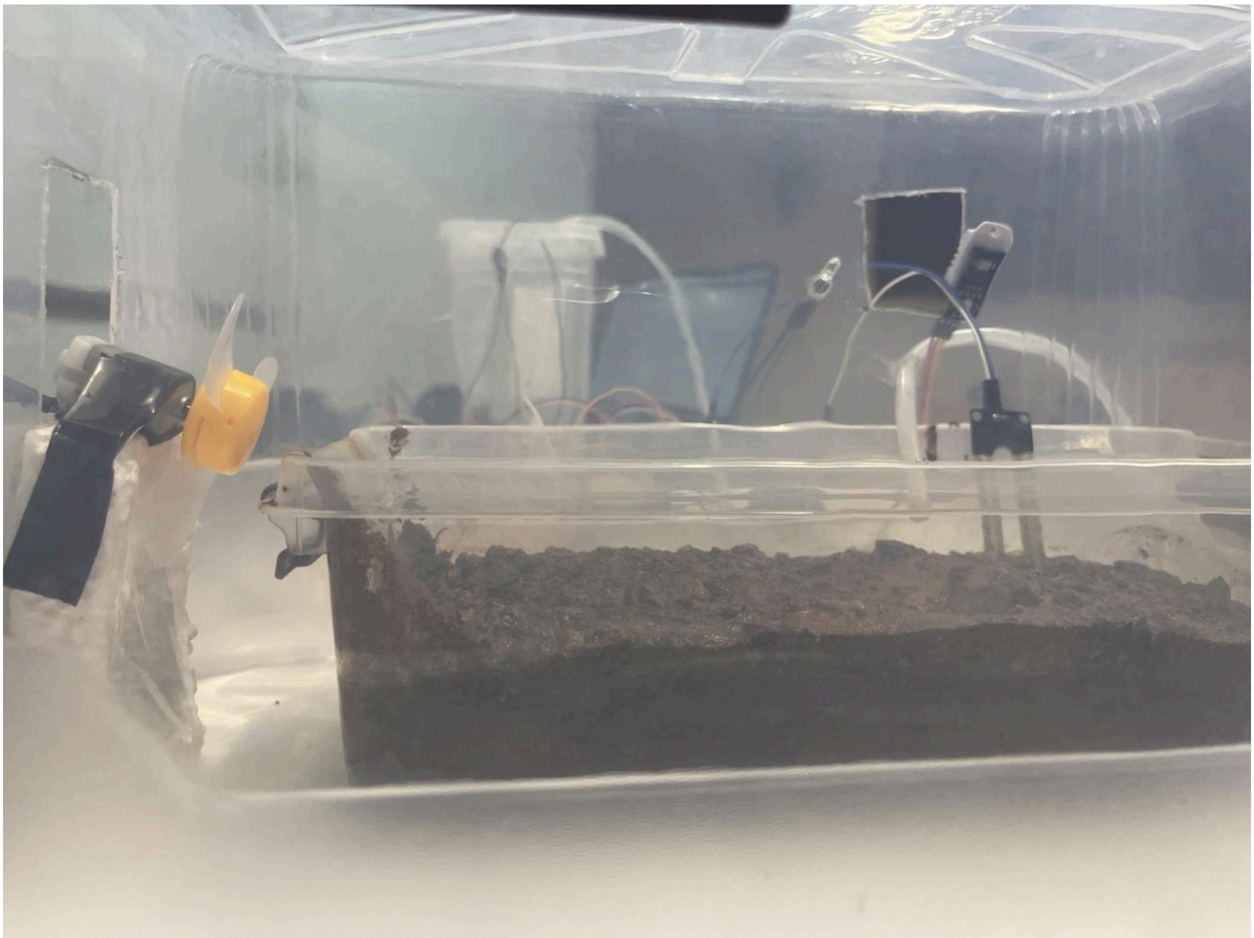


Figure 23. Front view of the greenhouse chamber showing the soil tray, fan and probes.

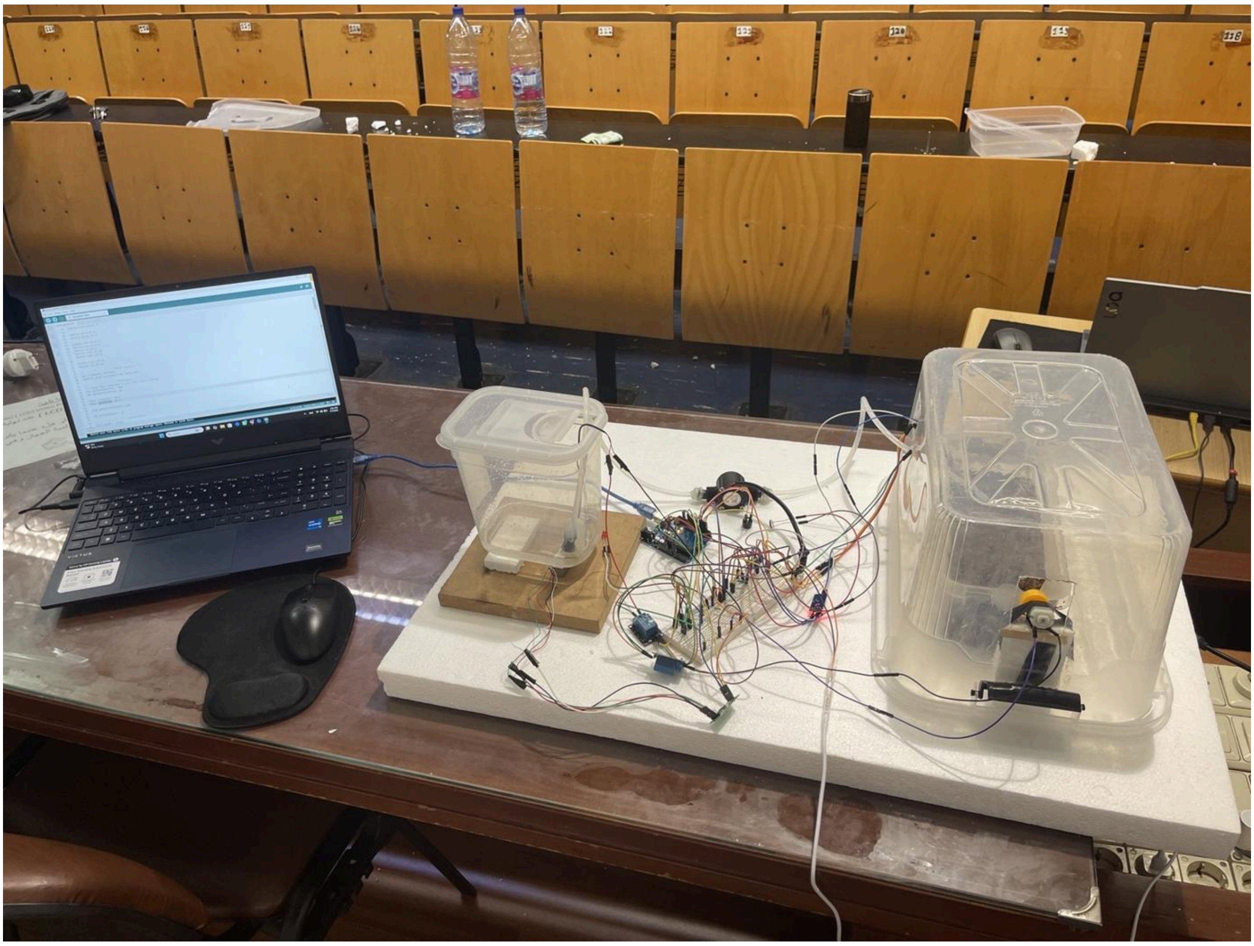


Figure 24. Top view of the complete prototype during laboratory integration.

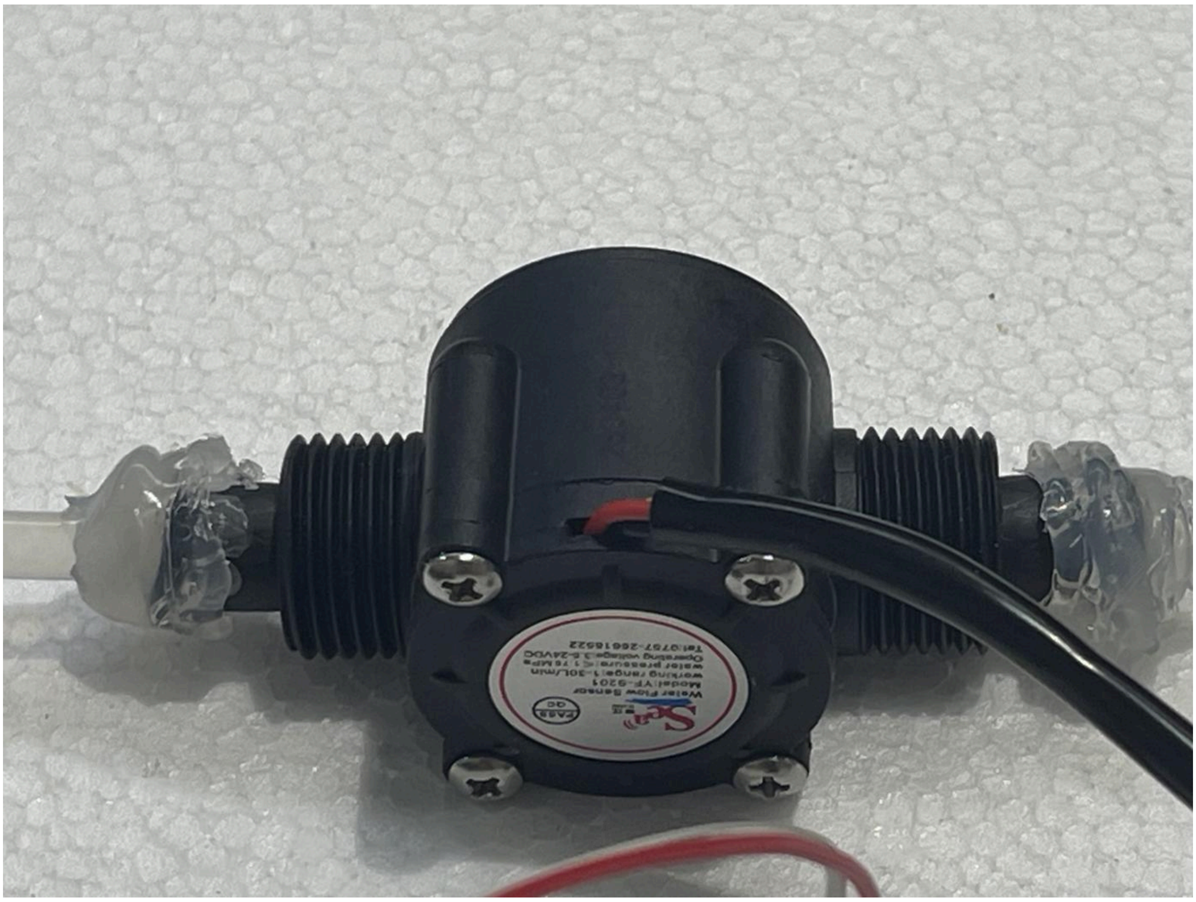


Figure 25. Close view of the installed black YF-S201 water-flow sensor.

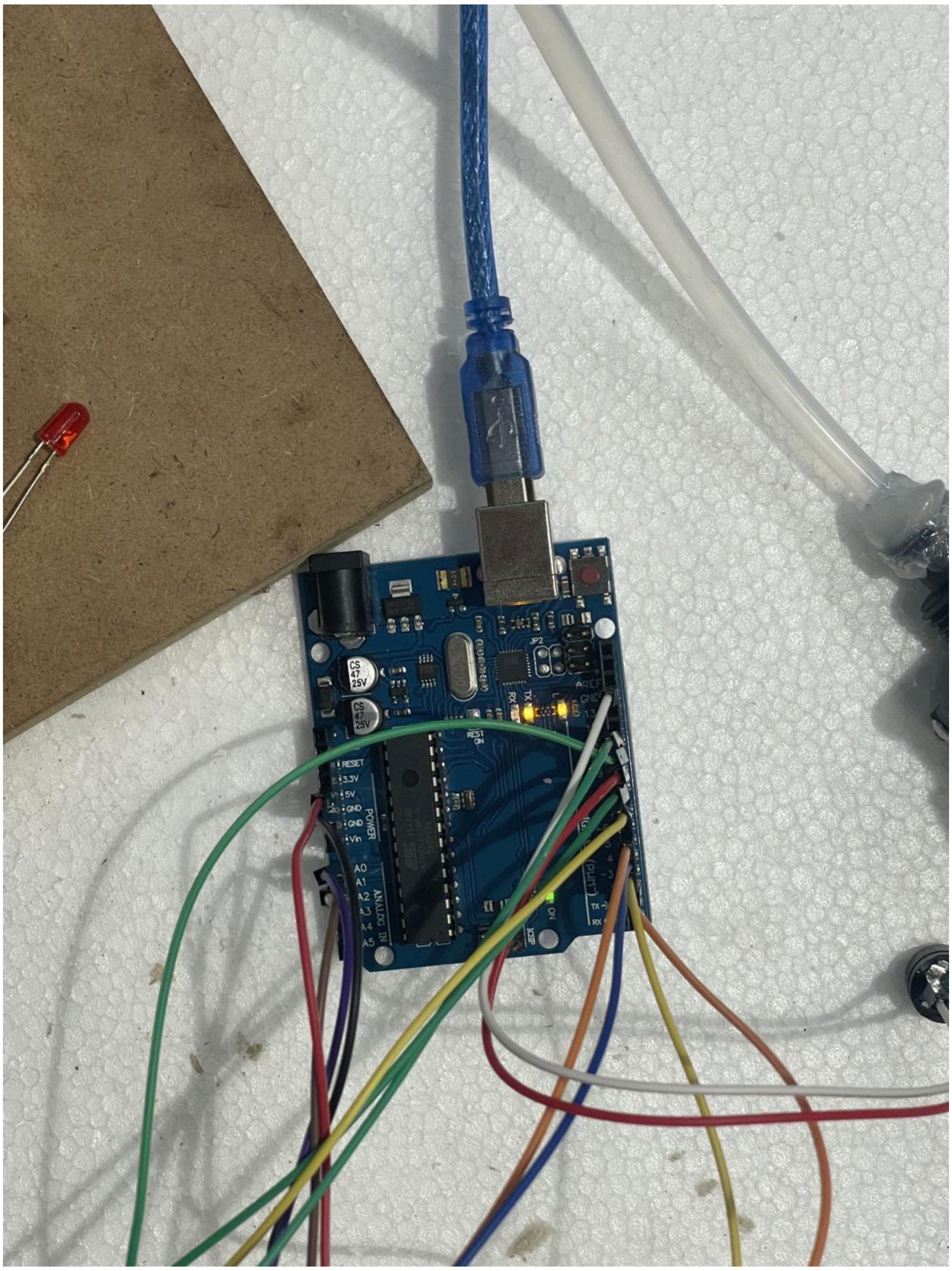


Figure 26. Arduino Uno controller and USB connection.

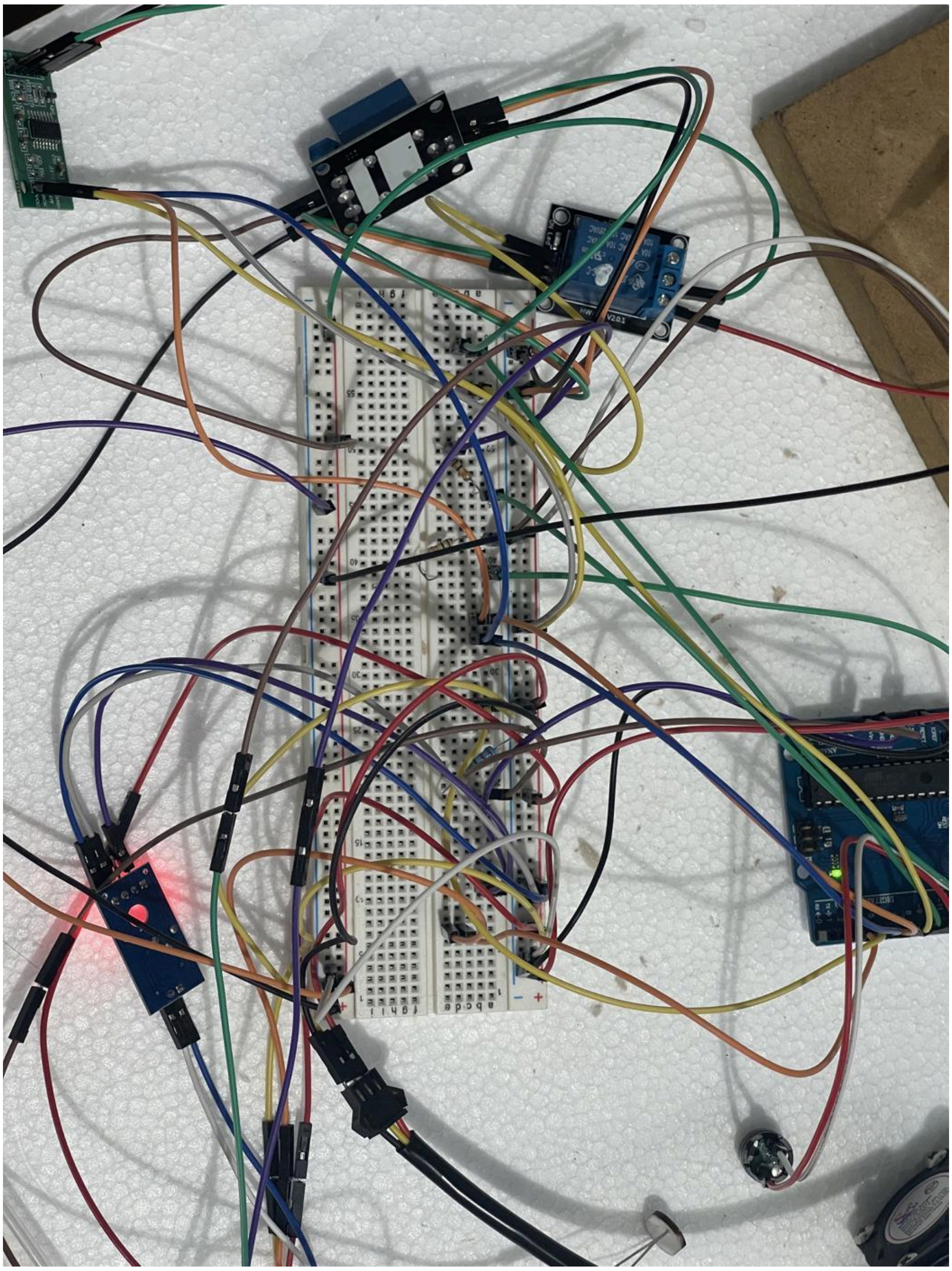


Figure 27. Breadboard wiring, relay module, HX711 board and connected sensors.